KNOWLEDGE 2 CONNECT

We are grateful to our sponsors:

Karger

Wolters Kluwer

EBSCO INFORMATION SERVICES

Ovid®

VZW VRIENDEN van de Biomedische Bibliotheek

ELSEVIER

UNIVERSITEIT GENT

RESEARCH DEPARTMENT
UNIVERSITY LIBRARY

KNOWLEDGE 2 CONNECT

RESEARCH DATA MANAGEMENT

DEPOSIT IN REPOSITORIES

UGent Data Stewards – June 2024

UNIVERSITEIT
GENT

This presentation includes slides reused from Elixir Belgium (CC BY 4.0)

OUTLINE

- Why? Which policies apply?
- Publishing data
- Identifiers
- Terms for reuse
- Understanding the data
- Q&A



PUBLICATIONS AND DATA

UNIVERSITEIT GENT

'Publications and Data' by Auke Herrema, licensed under CC-BY 4.0

OPEN SCIENCE

"Open science is an approach based on open cooperative work and systematic sharing of knowledge and tools as early and widely as possible in the process"

European Commission

- Open Access to publications
- Responsible management of data (FAIR principles)
- Open access to data ‘as open as possible, as closed as necessary’
- Information about outputs / tools / instruments to validate/re-use results and data
- Digital /physical access of results to validate the conclusions

CHANGING LEGAL & POLICY LANDSCAPE

Growing expectations of open & FAIR data



UNIVERSITEIT GENT

https://www.ugent.be/en/research/datamanagement/policies/overview.htm

OPEN SCIENCE IN HORIZON EUROPE



- Makes open science practices subject to evaluation at the proposal stage
- Introduces open science obligations for beneficiaries

Mandatory OS Practices

https://onderzoektips.ugent.be/en/tips/00001972/

Open Access to publications

Research Data Management (RDM)

-> Asap deposit in trusted repository

Recommended OS Practices

https://onderzoektips.ugent.be/en/tips/00001973/

Early and open sharing of research

Preregistration, open peer-review

Citizen science, society engagement

Research output management (beyond data)

Reproducible outputs

UNIVERSITEIT
GENT

CLINICAL DATA SHARING

Clinical Trials

Clinical trial regulation: from 31 January 2022

publish all clinical trial-related information generated during the life cycle of a clinical trial

- Protocol
- Assessment and decision on trial conduct
- Summary of trial results including a lay summary
- Study reports for those trials in the system subsequently included in a marketing authorisation submission in the EU
- Inspections
- etc.

Through the Clinical Trials website https://euclinicaltrials.eu/

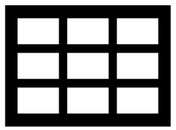
UNIVERSITEIT GENT

NEW FWO REQUIREMENTS

- DMP template: the Flemish DMP template
- Reporting requirements
  - Evaluation of scientific reports based on FRIS data
- Mapping the output of research projects
  - https://www.fwo.be/en/about-fwo/research-policy/research-results/



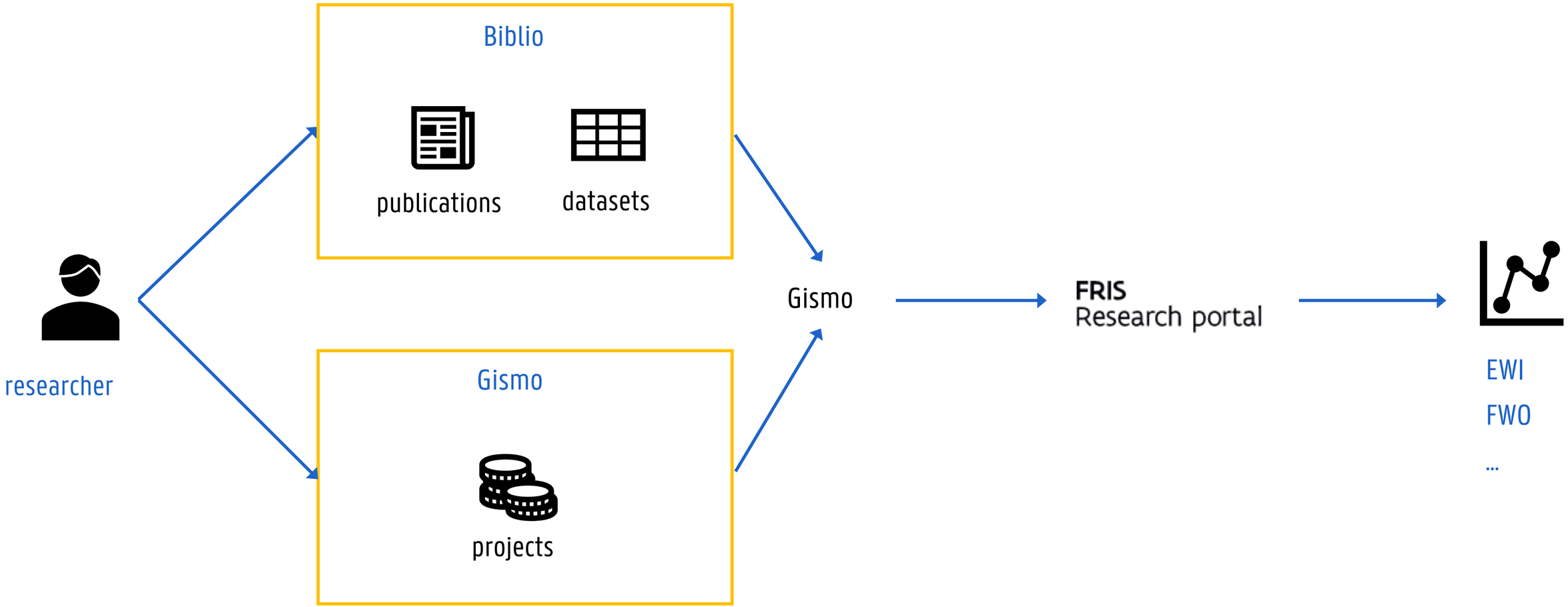
publications



datasets

https://onderzoektips.ugent.be/nl/tips/00001694/

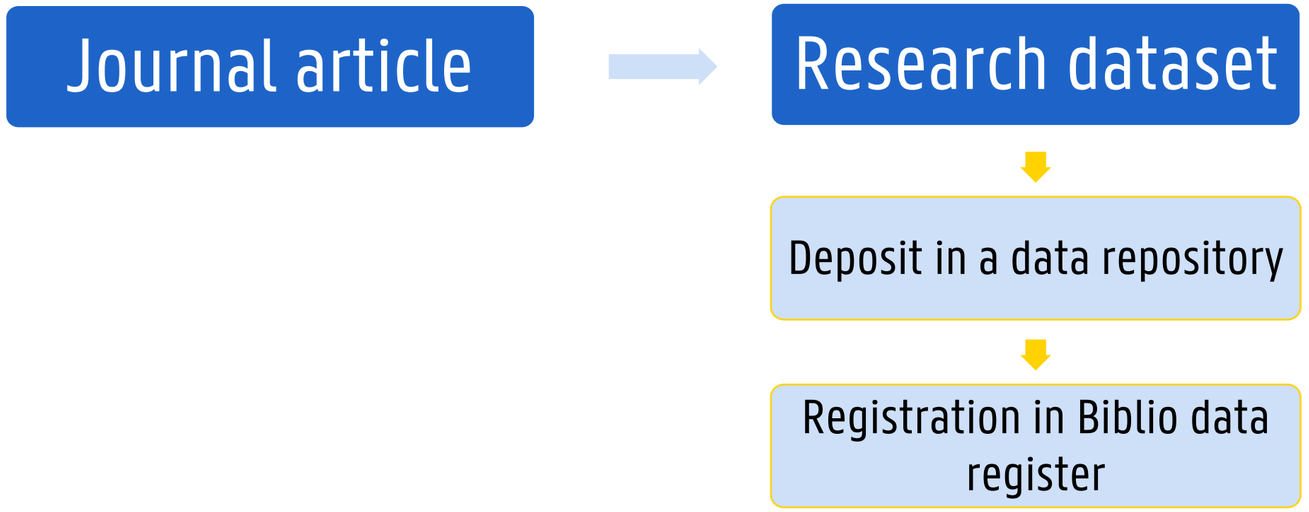
REPORTING TO FLEMISH RESEARCH PORTAL (FRIS)



https://researchportal.be/en

UGENT POLICY ON SCHOLARLY PUBLISHING (AS OF 2023)

- Mandatory **registration** of published research datasets, i.e. entering and/or validating the required bibliographic details ('metadata') of your published research output.
- Research datasets published via a data repository, with at least one Ghent University creator at the time of publication, and which support publications from 2023 or later.

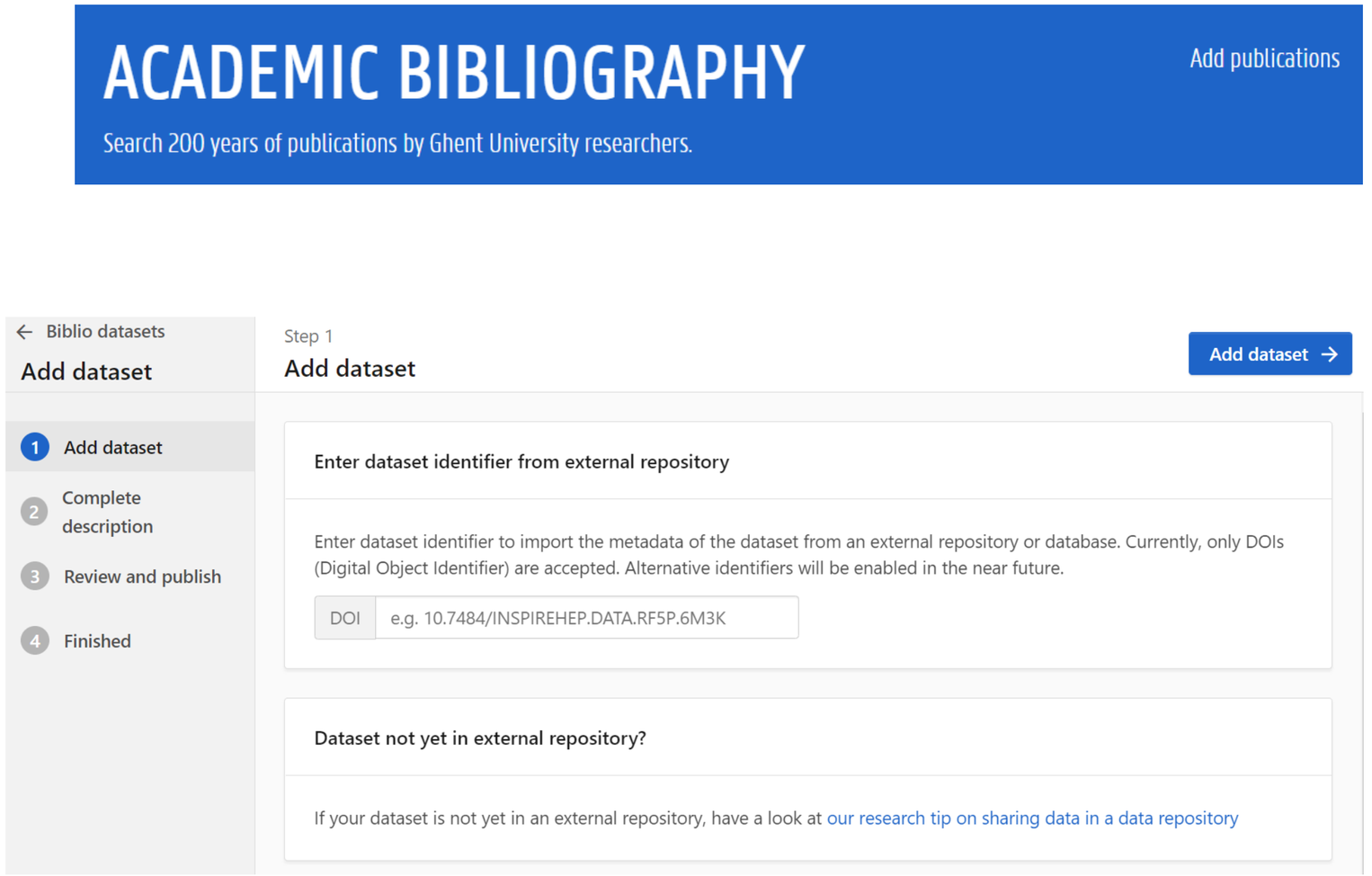


UNIVERSITEIT GENT

https://www.ugent.be/intranet/en/research/impact/schol-publishing/policy-ugent.htm

BIBLIO: DATASETS AND SOFTWARE

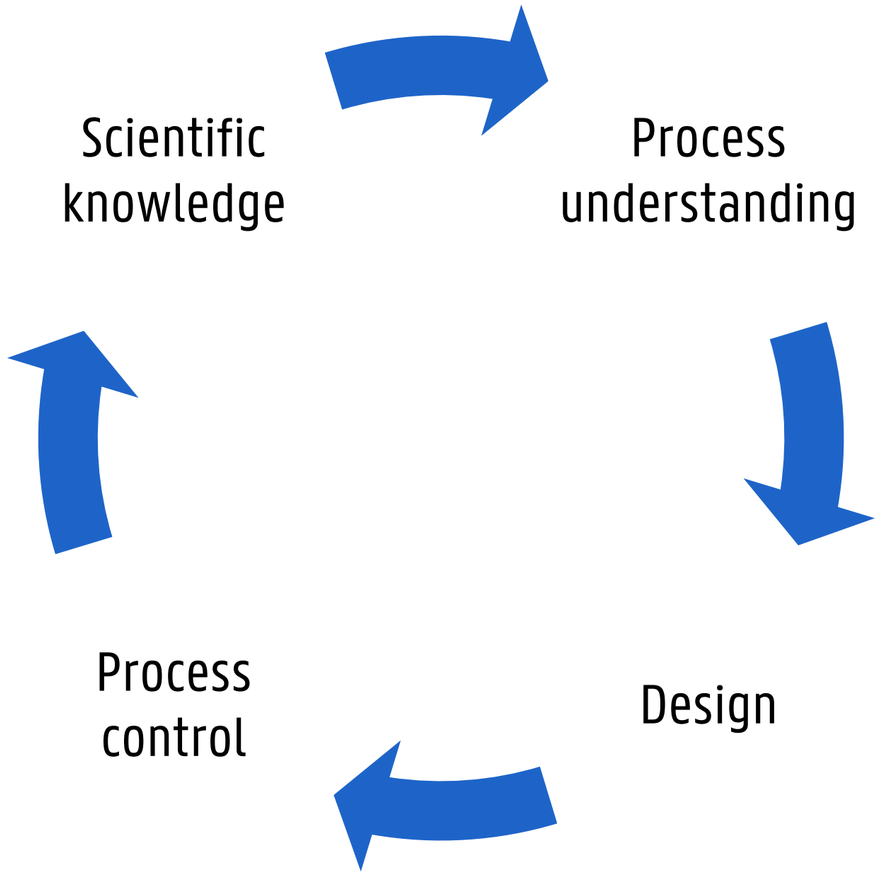
- Register datasets in Biblio using DOI
- Automatically harvests several metadata fields
- Mandatory with new UGent policy
- Enhance findability of your research outputs



UNIVERSITEIT GENT

PLANNING IS ESSENTIAL

To avoid problems in the long run…

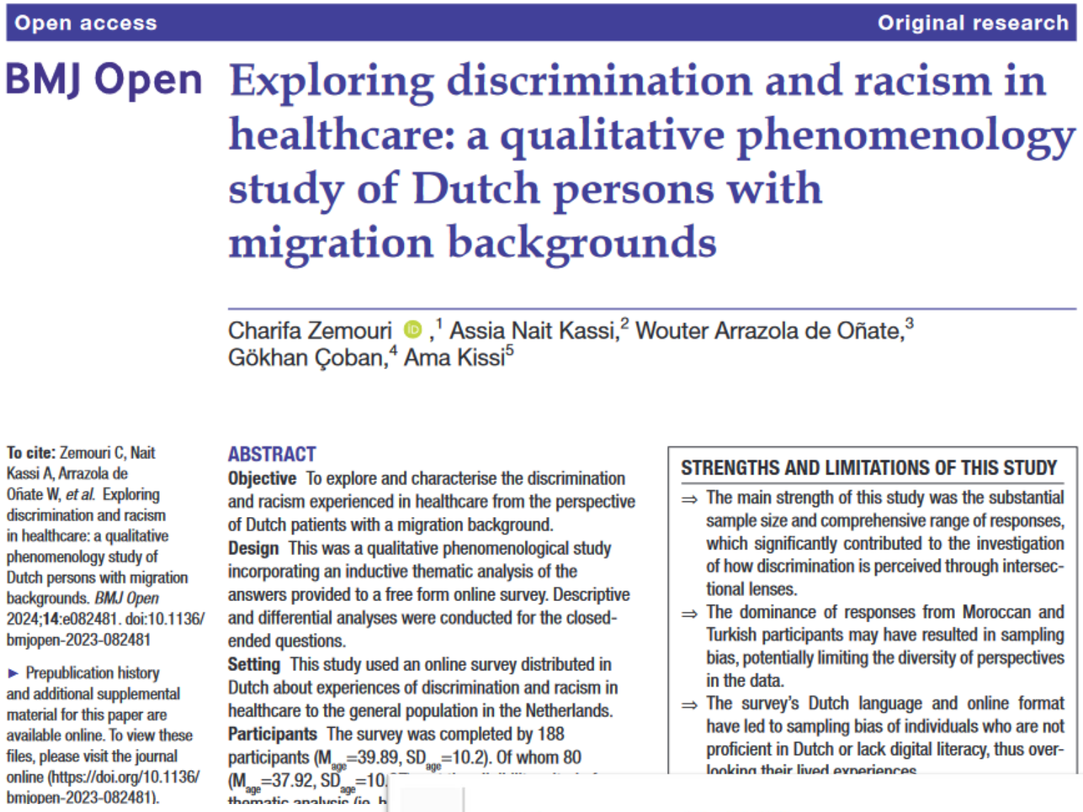


DMP ONLINE .BE

UNIVERSITEIT GENT

KEEP THE END IN MIND

E.g. http://doi.org/10.1136/bmjopen-2023-082481



Open access Original research

BMJ Open

Exploring discrimination and racism in healthcare: a qualitative phenomenology study of Dutch persons with migration backgrounds

Charifa Zemouri ,[1] Assia Nait Kassi,[2] Wouter Arrazola de Oñate,[3] Gökhan Çoban,[4] Ama Kissi[5]

To cite: Zemouri C, Nait Kassi A, Arrazola de Oñate W, *et al*. Exploring discrimination and racism in healthcare: a qualitative phenomenology study of Dutch persons with migration backgrounds. *BMJ Open* 2024;**14**:e082481. doi:10.1136/bmjopen-2023-082481

► Prepublication history and additional supplemental material for this paper are available online. To view these files, please visit the journal online (https://doi.org/10.1136/bmjopen-2023-082481).

ABSTRACT

Objective To explore and characterise the discrimination and racism experienced in healthcare from the perspective of Dutch patients with a migration background.

Design This was a qualitative phenomenological study incorporating an inductive thematic analysis of the answers provided to a free form online survey. Descriptive and differential analyses were conducted for the closed-ended questions.

Setting This study used an online survey distributed in Dutch about experiences of discrimination and racism in healthcare to the general population in the Netherlands.

Participants The survey was completed by 188 participants (M_{age}=39.89, SD_{age}=10.2). Of whom 80 (M_{age}=37.92, SD_{age}=10.

thematic analysis (o

STRENGTHS AND LIMITATIONS OF THIS STUDY

⇒ The main strength of this study was the substantial sample size and comprehensive range of responses, which significantly contributed to the investigation of how discrimination is perceived through intersectional lenses.

⇒ The dominance of responses from Moroccan and Turkish participants may have resulted in sampling bias, potentially limiting the diversity of perspectives in the data.

⇒ The survey's Dutch language and online format have led to sampling bias of individuals who are not proficient in Dutch or lack digital literacy, thus overlooking their lived experiences

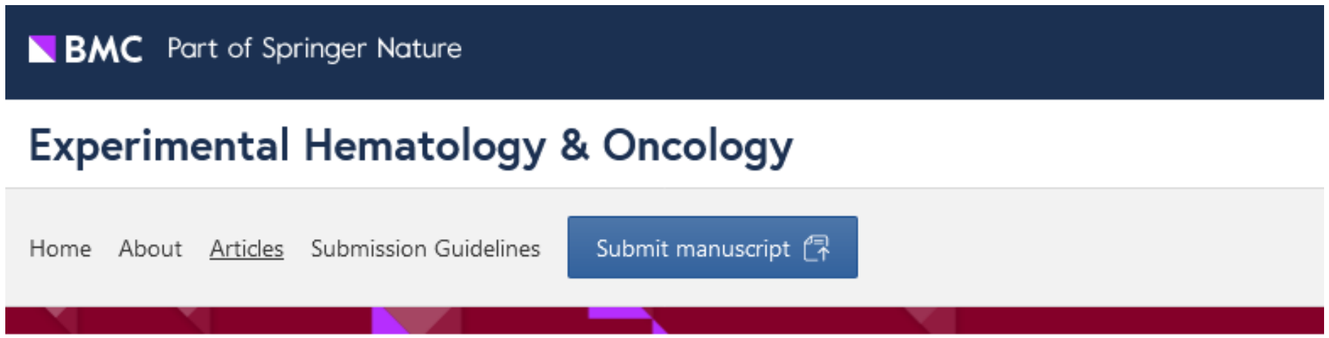
Data availability statement

All data relevant to the study are included in the article or uploaded as supplementary information.

UNIVERSITEIT GENT

KEEP THE END IN MIND (2)

E.g. https://doi.org/10.1186/s40164-024-00524-4



Research | Open access | Published: 03 June 2024

High efficacy of huCD20-targeted AcTaferon in humanized patient derived xenograft models of aggressive B cell lymphoma

Willem Daneels, Alexander Van Parys, Leander Huyghe, Elke Rogge, Steffi De Rouck, Ruben Christiaen, Lennart Zabeau, Sylvie Taveirne, Jo Van Dorpe, Niko Kley, Anje Cauwels, Erik Depla, Jan Tavernier & Fritz Offner

Experimental Hematology & Oncology **13**, Article nu

177 Accesses | **6** Altmetric | Metrics

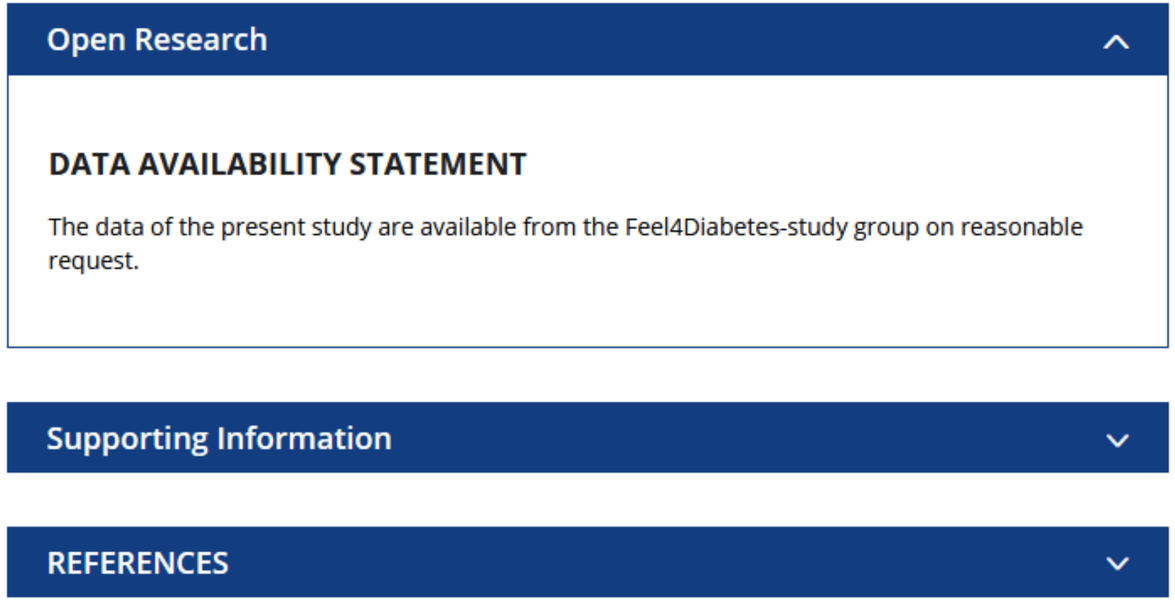
Availability of data and materials

All data supporting the findings of this study are available within the paper, RNA-Seq data for DFBL-98848-V3-mCLP (Proxe_MCL) is available at https://www.ncbi.nlm.nih.gov/sra/SAMN06621890, any additional data are available from the corresponding author upon reasonable request.

UNIVERSITEIT GENT

DATA UPON REQUEST

- Data available upon request increases the chances that data will be unavailable.
- "Previous research has found that **as little as 11% of data was readily made available upon request**, with the likelihood of data being available rapidly declining over time[1]"
  - → original author could not be reached
  - → refused to share, or
  - → data were lost or unavailable
- More recently, "**only 6,8% provided the requested data**[2]"



UNIVERSITEIT GENT

1 Federer, Lisa M et al. "Data sharing in PLOS ONE: An analysis of Data Availability Statements." PloS one vol. 13,5 e0194768. 2 May. 2018, doi:10.1371/journal.pone.0194768

2 Gabelica et al. 2022, https://doi.org/10.1016/j.jclinepi.2022.05.019

WAYS OF SHARING ('PUBLISHING') DATA



Upon request



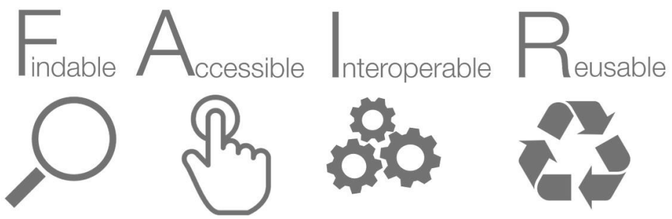
Personal or project website



Article suppl. material



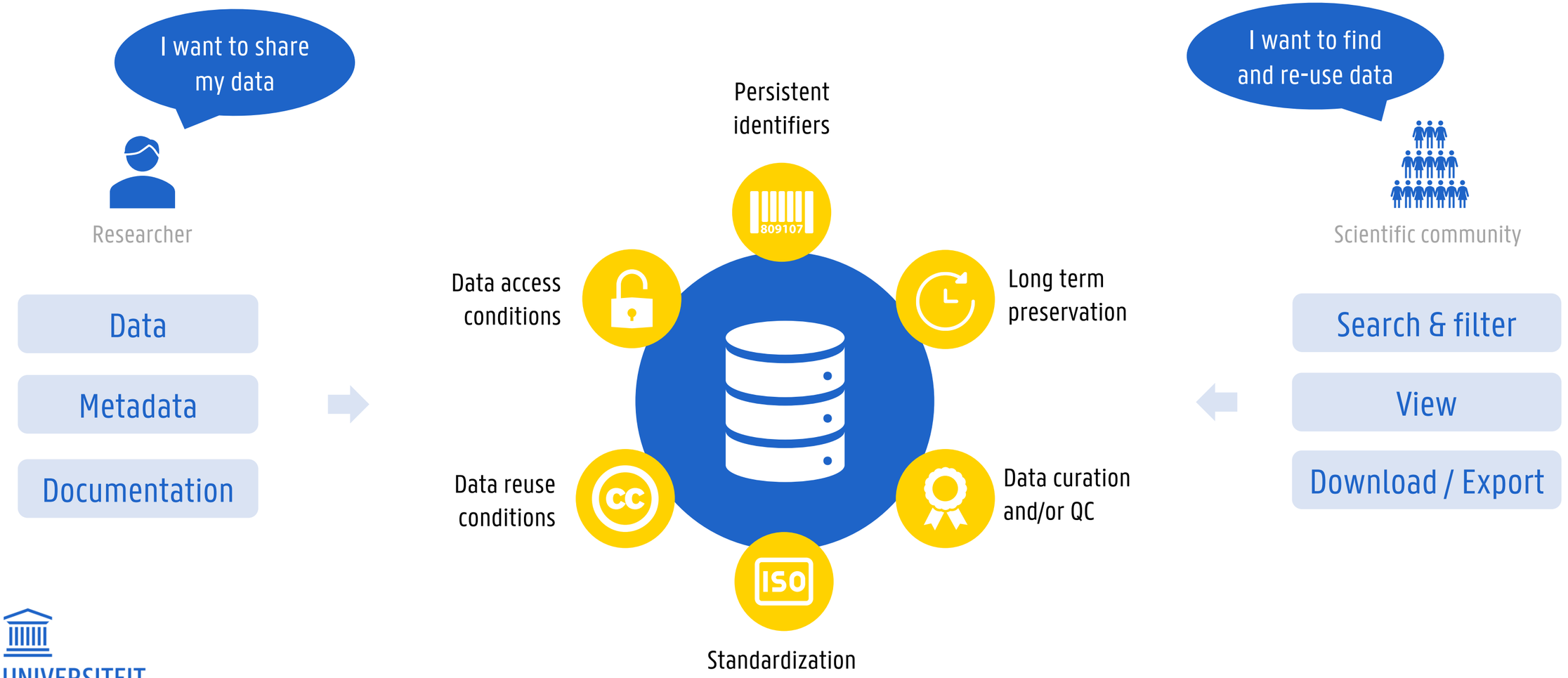
Data repository or archive



UNIVERSITEIT GENT

DEPOSIT THE DATA IN A TRUSTED REPOSITORY

A database infrastructure that collects, manages and stores data for preservation and sharing



UNIVERSITEIT GENT

SOME TERMINOLOGY

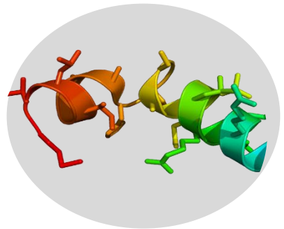
UNIVERSITEIT
GENT

WHAT IS RESEARCH DATA? WHAT IS RESEARCH OUTPUT?

Survey



Measurement



Observation



Code



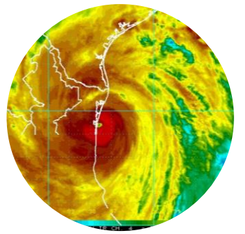
Image



Document



Visualisation



Data as well as other research outputs that may be generated or reused throughout their projects, either
digital (e.g. software, workflows, protocols, models, etc.) or
physical (e.g. new materials, antibodies, reagents, samples, etc.).

UNIVERSITEIT GENT

FAIR DATA PRINCIPLES

Findable → How easily can humans (and machines) find my data? How good are my metadata?

Accessible → Once found, how easy is it to access my data?
Are the metadata accessible, even if the data aren't?

Interoperable → How well does my data integrate with other data? Is a standard vocabulary used?

Reusable → How difficult is it to reuse my data? Are the (meta)data well described? Is there a usage license?

UNIVERSITEIT GENT

FINDABLE IDENTIFIERS FOR DATA

UNIVERSITEIT GENT

TO REPORT/CITE YOU NEED TO BE ABLE TO REFER

- Identifier needed to refer to dataset

Examples:

- Clinical trials ClinicalTrials.gov NCT04330638
- ENA project https://www.ebi.ac.uk/ena/browser/view/PRJEB39409
- Biostudies / ArrayExpress: https://www.ebi.ac.uk/biostudies/arrayexpress/studies/E-MTAB-13427
- GEO accession number: https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE207592
- …

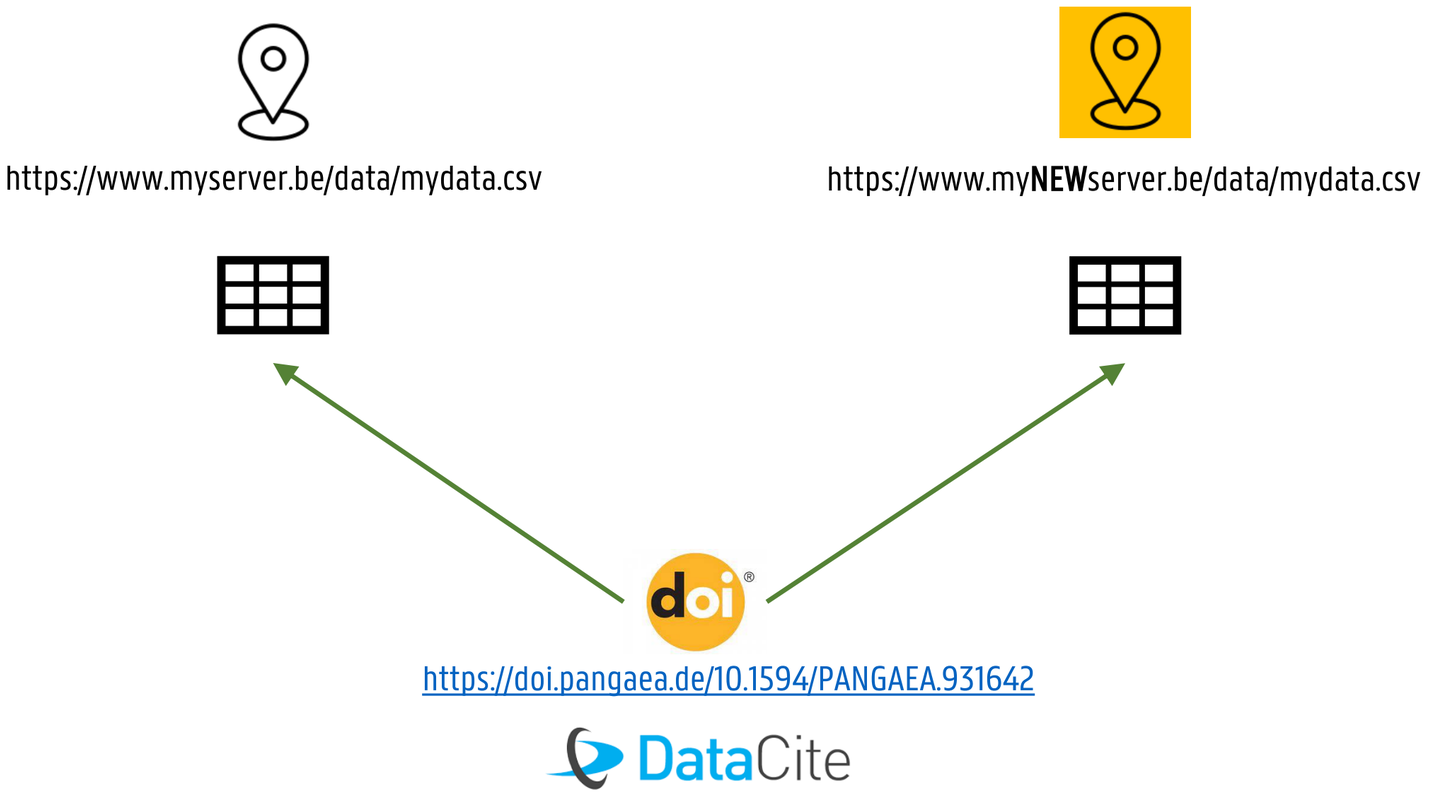
Though preferably you want an identifier that is persistent in time and space

A TRUSTED REPOSITORY WILL PROVIDE A PERSISTENT IDENTIFIER

GLOBALLY UNIQUE IDENTIFIER:
ensures that there are no two identical identifiers that point to different digital objects

PERSISTENT IDENTIFIER:
ensures that the digital object will remain findable over time and reduces the risk of broken links

Example: DOI - digital object identifier



UNIVERSITEIT GENT

PERSISTENT IDENTIFIERS: EXAMPLES

Examples of globally unique, persistent identifiers

Identifier:			
Organisation:			
Scope:	Digital objects	Researchers	Research organisations

What are persistent identifiers and when to use them: https://www.youtube.com/watch?v=iea6d5oI8Ag

UNIVERSITEIT GENT

Cross referencing resources

The benefit of depositing your data

Europe PMC

Abstract
Figures (8)
Free full text
Citations & impact
Data
Similar Articles
Funding

Stellate Cells, Hepatocytes, and Endothelial Cells Imprint the Kupffer Cell Identity on Monocytes Colonizing the Liver Macrophage Niche.

Bonnardel J [1], T'Jonck W [1], T'Jonck W [1], Gaublomme D [2], Browaeys R [3], Scott CL [4], Martens L [5], Vanneste B [1], De Prijck S [1], Nedospasov SA [6], Kremer A [7], Van Hamme E [7], Borghgraef P [7], Toussaint W [8], De Bleser P [5], Mannaerts I [9], Beschin A [10], van Grunsven LA [9], Lambrecht BN [8], Taghon T [11] ... [Show all 24] ... Guilliams M [1]

Author information

Immunity, 24 Sep 2019, 51(4):638-654.e9
DOI: 10.1016/j.immuni.2019.08.017 PMID: 31561945 PMCID: PMC6876284
Free to read & use

A comment on this article appears in "Kupffer cells: finding their niche." Nat Rev Immunol. 2019 Nov;19(11):660-661.

Annotating publications with the related data

UNIVERSITEIT GENT

Data

Data behind the article
This data has been text mined from the article, or deposited into data resources.

BioStudies: supplemental material and supporting data
http://www.ebi.ac.uk/biostudies/studies/S-EPMC6876284?xr=true

Genes & Proteins
UniProt - P36970 (1 citation)

GEO - Gene Expression Omnibus
GEO - GSE135790 (3 citations)

Nucleotide Sequences
ENA - C10634 (1 citation)

Data that cites the article
This data has been provided by curated databases and other sources that have cited the article.

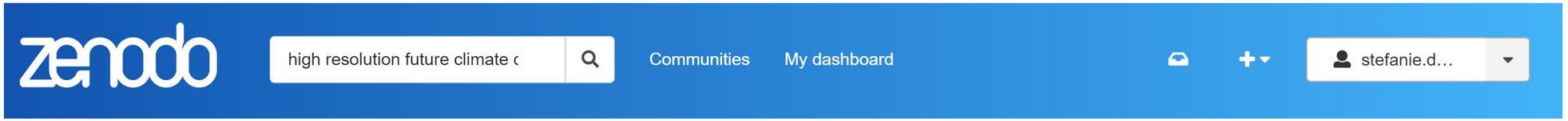
Nucleotide Sequences (4)
Stellate cells, hepatocytes and endothelial cells imprint the Kupffer cell identity on monocytes colonizin... (ENA - PRJNA560060)
Stellate cells, hepatocytes and endothelial cells imprint the Kupffer cell identity on monocytes colonizin... (ENA - PRJNA560063)
Stellate cells, hepatocytes and endothelial cells imprint the Kupffer cell identity on monocytes colonizin... (ENA - PRJNA560062)
Sequence Read Archive(SRA) (ENA - SRP218244)

Genes & Proteins (Showing 5 of 12)
Tumor necrosis factor (UniProt - A0A0U5J4W4)
Tumor necrosis factor (UniProt - P06804)

ACCESSIBLE PUBLISHING DATA = DATA SHARING

UNIVERSITEIT GENT

ACCESSIBLE ≠ OPEN



April 30, 2020 (v1) | Dataset | Open

High-resolution future climate data for species distribution models in Europe

De Troch, Rozemien; Termonia, Piet; Van Schaeybroeck, Bert

Description This dataset contains a set of 13 climatological variables (Variable, VariableName) at a spatial resolution of 1x1km for Europe (nx = 13147, ny = 6071) for historical (ClimatePeriod) and future climate conditions. These variables are a subset of the so-called bioclimatic variables that are often part of global gridded datasets (e.g...

Uploaded on April 30, 2020

683 623

March 13, 2023 (v1) | Dataset | Embargoed

Flemish Political Ambition Survey

Bram Wauters; Robin Devroe; Audrey Vandeleene

Dataset based on a survey about political ambition among a random sample of the youth population (aged 18-35) in Flanders (Belgium), N = 1,000

Uploaded on March 13, 2023

32 0

November 12, 2018 (v3) | Dataset | Restricted

Data for development, evaluation and application of NicheNet

Browaeys, Robin; Saelens, Wouter; Saeys, Yvan

Here you can find the datasets used for development, validation and application of NicheNet, a method that links ligands to target genes to model intercellular communictation. The software to use NicheNet (i.e. R package nichenetr) is available at: https://github.com/browaeysrobin/nichenetr Short description of individual files: * data for d...

Uploaded on November 12, 2018

2 more versions exist for this record

425 6

UNIVERSITEIT GENT

IS ZENODO SUITABLE FOR RESTRICTED ACCESS?

- **Content scope**: All fields of research. All types of research artifacts. Content must not violate privacy or copyright, or breach confidentiality or non-disclosure agreements for data collected from human subjects.
- Zenodo is an open dissemination research data repository, and the uploader of content responsible to ensure that the content is suitable for open dissemination and that it complies with applicable laws, including, but not limited to, privacy, data protection, and intellectual property rights. **Thus**, an uploader shall ensure that sensitive personal data is either anonymized to an appropriate degree or fully consent cleared. Uploaders considering Zenodo for the storage of non anonymized or encrypted/unencrypted sensitive personal data are therefore advised to use bespoke platforms rather than open dissemination services like Zenodo for sharing their data.

https://help.zenodo.org/faq/
https://about.zenodo.org/policies/
https://about.zenodo.org/terms/

RESTRICTIONS ON DATA/SOFTWARE SHARING



Personal data



Confidential data



Sensitive data



Third-party data



Copyright



Valorization

UNIVERSITEIT GENT

More info at: https://www.ugent.be/en/research/datamanagement/after-research/sharing.htm#Restrictionsondatasharing

WHAT ABOUT SHARING OF SENSITIVE OR CONFIDENTIAL DATA?

"As open as possible, as closed as necessary"



Can be freely used, modified & shared by anyone for any purpose

Can be used under conditions, with restrictions, after approval

Under embargo
Unable to share

QUESTION: where does each of your datasets fit?

UNIVERSITEIT GENT

WHAT ABOUT SHARING OF SENSITIVE DATA?

1. Make sure approvals are in place: ethical and legal
2. Use a repository that
   - enables findability
   - protects from unauthorized access
   - has compatible terms and conditions

e.g. The European Genome-phenome Archive (EGA) is a service for permanent archiving and sharing of personally identifiable genetic, phenotypic, and clinical data generated for the purposes of biomedical research projects

EUROPEAN GENOME-PHENOME ARCHIVE

Central metadata accessibility

Secure storage

Implemented data access committees

or DANS-EASY or UK Data Archive, …

UNIVERSITEIT GENT

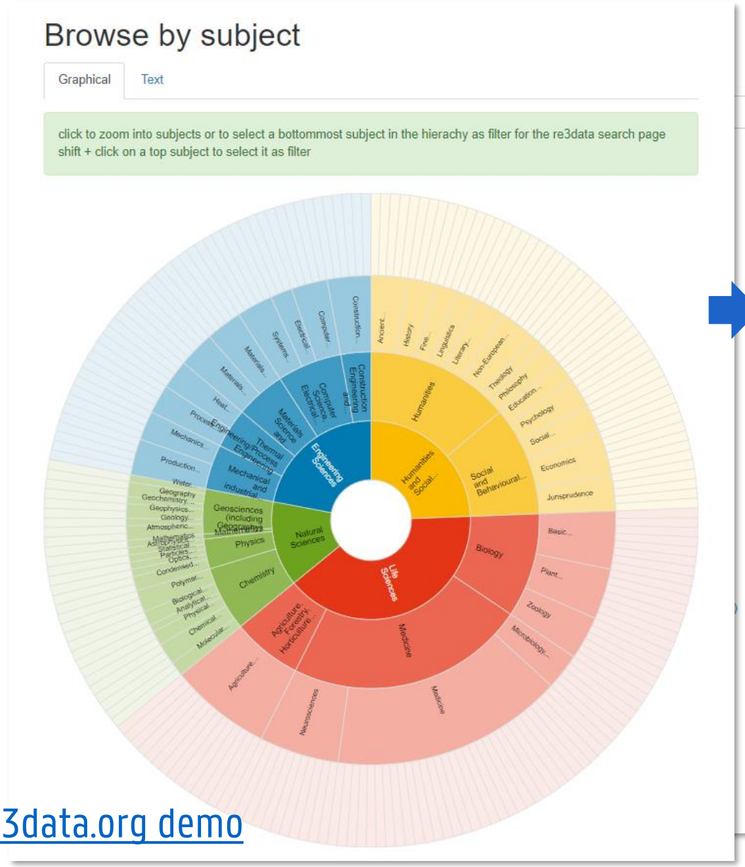
PROJECT DATASETS (EXAMPLE)

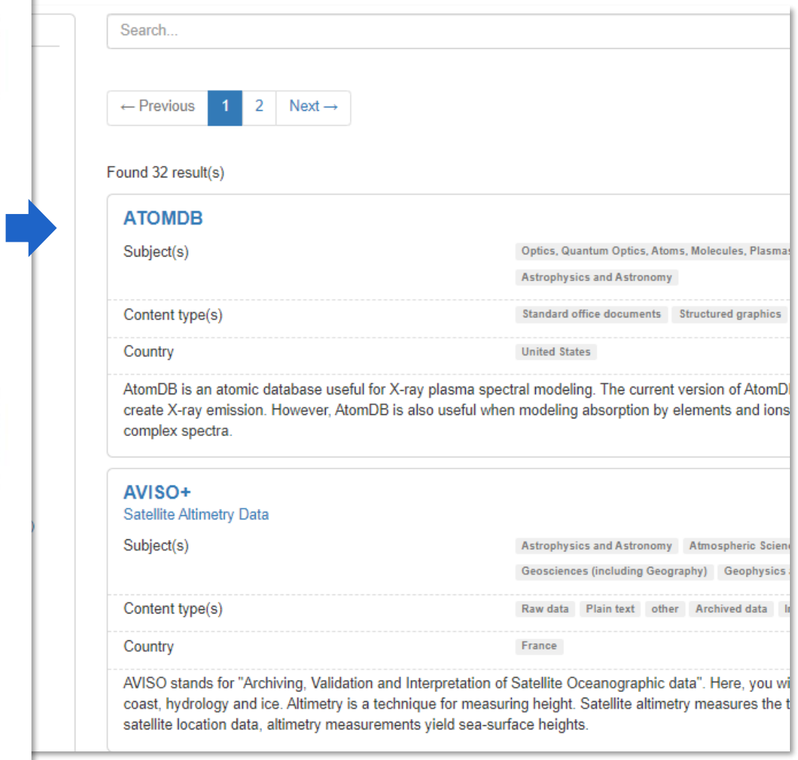
Preclinical: gene expression in cell lines. Confirmation based on tissue samples of patients.

Dataset	Description	Sharing	Repository
Cell line expression values	Experimental: qPCR gene expression measurement on treated cell cultures	Yes, open	Biosamples
eCRF clinical samples parameters	Minimal set of demographic and diagnostic parameters about patients and their samples	No, unless further processed to complete anonymity	N/A
Tissue expression values	Experimental: qPCR gene expression measurement on patient samples	Restricted, because link to patient ID is maintained (pseudonymous)	EGA

HOW TO SELECT A DATA REPOSITORY?

1. Where do your peers publish data? Domain's consensus
2. Does the journal / funder recommend or require certain repositories?
3. Registries of data repositories: re3data.org, FAIRsharing





re3data.org demo

Data Repository Guidance

Scientific Data mandates the release of datasets accompanying our Data Descriptors.

View data repositories

- **Biological sciences:** Nucleic acid sequence; Protein sequence; Molecular & supramolecular structure; Neuroscience; Omics; Taxonomy & species diversity; Mathematical & modelling resources; Cytometry and Immunology; Imaging; Organism-focused resources
- **Health sciences**
- **Chemistry and Chemical biology**
- **Earth, Environmental and Space sciences:** Broad scope Earth & environmental sciences; Astronomy & planetary sciences; Biogeochemistry and Geochemistry; Climate sciences; Ecology; Geomagnetism & Palaeomagnetism; Ocean sciences; Solid Earth sciences
- **Physics**
- **Materials science**
- **Social sciences**
- **Generalist repositories**

Data repositories recommended by Nature

MORE POINTERS TO FIND REPOSITORY/IES FOR YOUR DATA

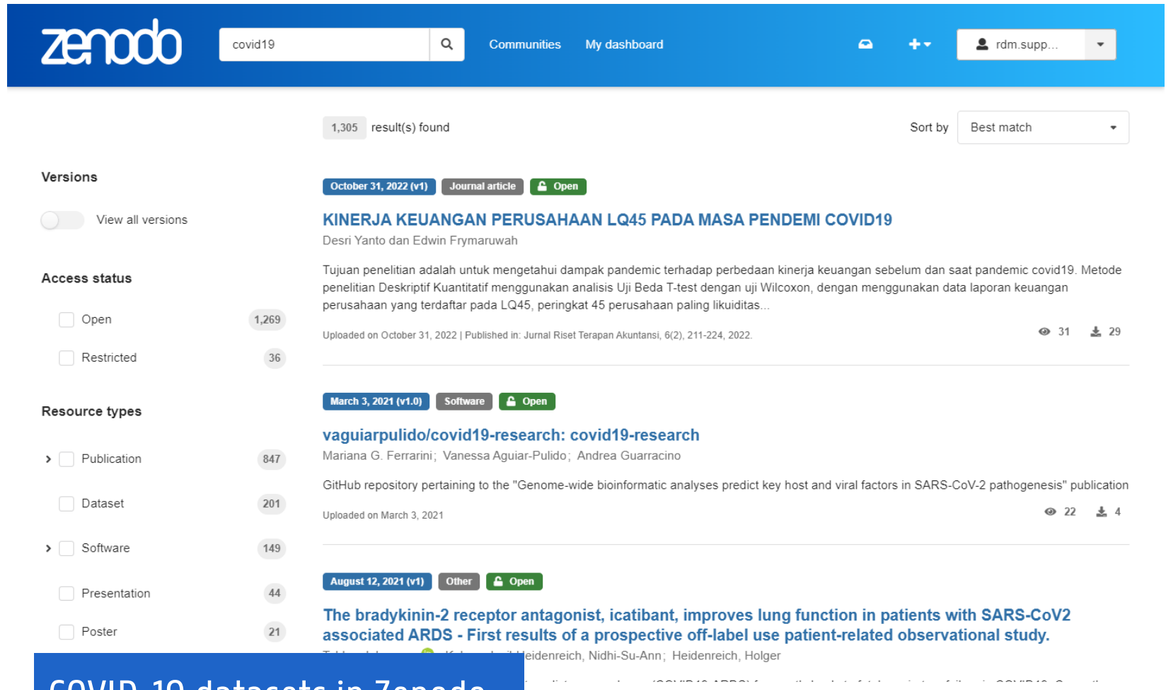
Where would your data fit?

Find a repository

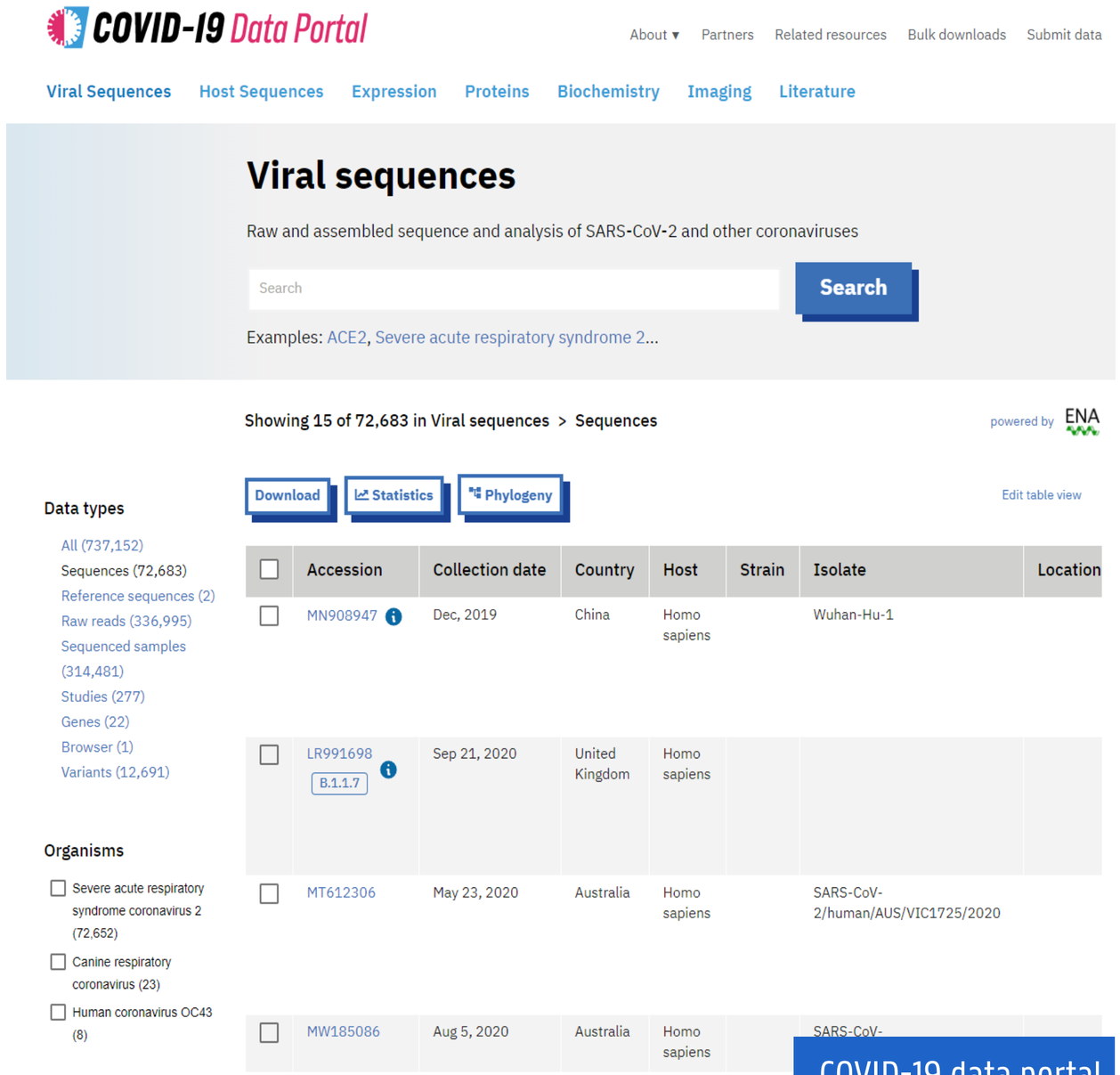
- EBI Submission wizard: https://www.ebi.ac.uk/submission/
- Biomolecular data: https://elixir-europe.org/platforms/data/elixir-deposition-databases
- https://re3data.org
  - Also use this register to validate repositories you find through search engines
- Data repositories recommended by NaturePG
- NIH supported data sharing resources: https://www.nlm.nih.gov/NIHbmic/domain_specific_repositories.html

UNIVERSITEIT GENT

GENERIC VS. DOMAIN-SPECIFIC REPOSITORIES



COVID-19 datasets in Zenodo

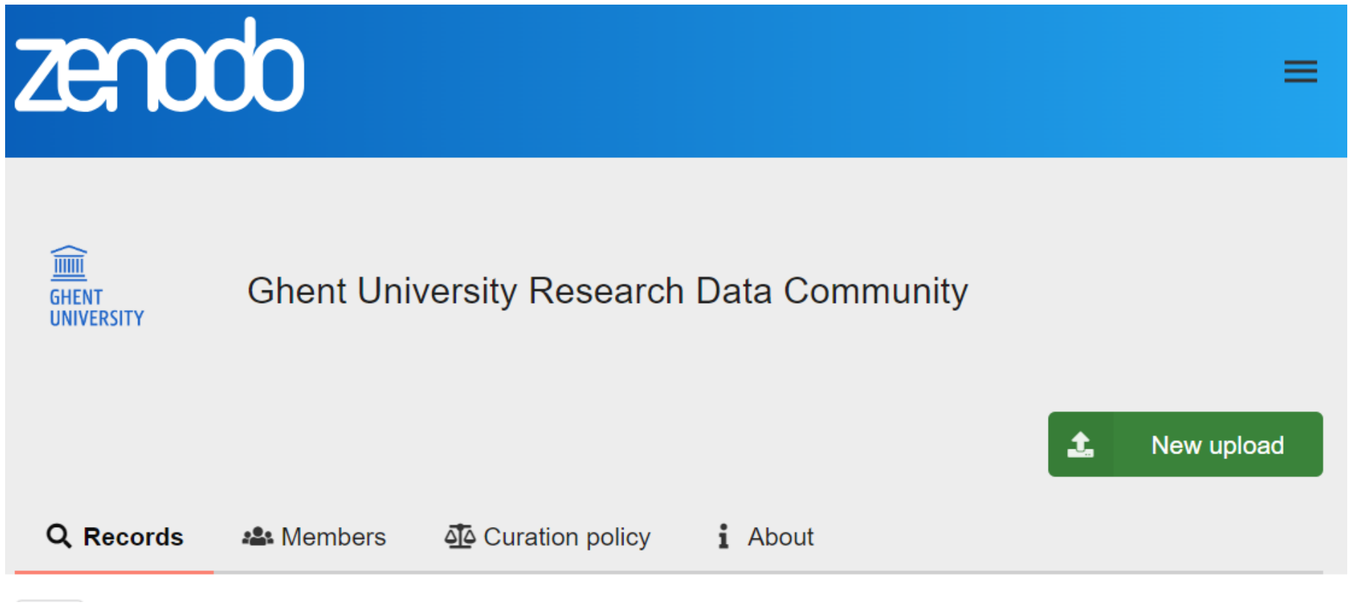


COVID-19 data portal

UNIVERSITEIT GENT

GENERAL PURPOSE REPOSITORY: ZENODO, WITH UGENT COMMUNITY

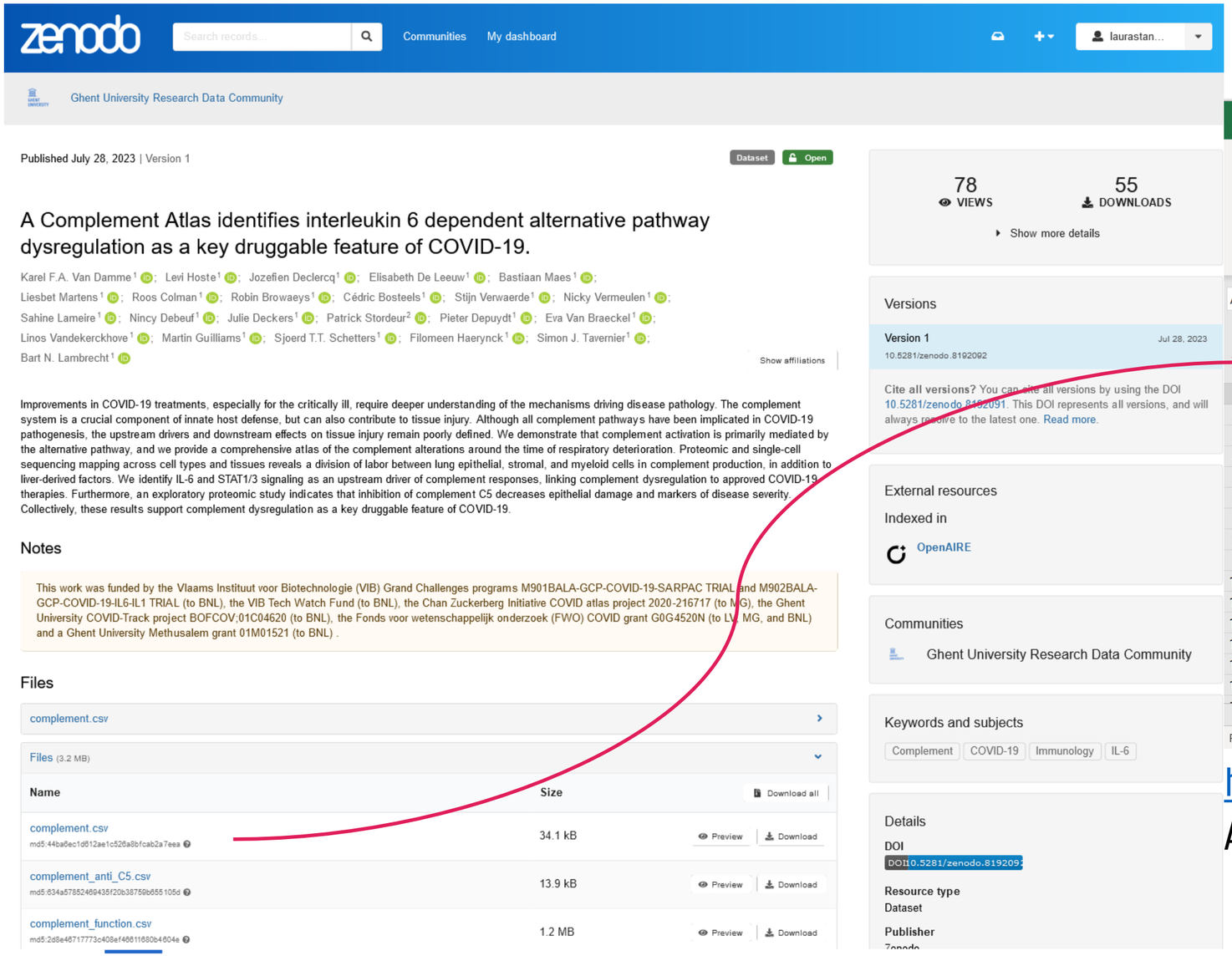
- ‘For the long tail of science’
- Curation by UGent Data Curators
- Accepts research data and software
- Avoid spam filters

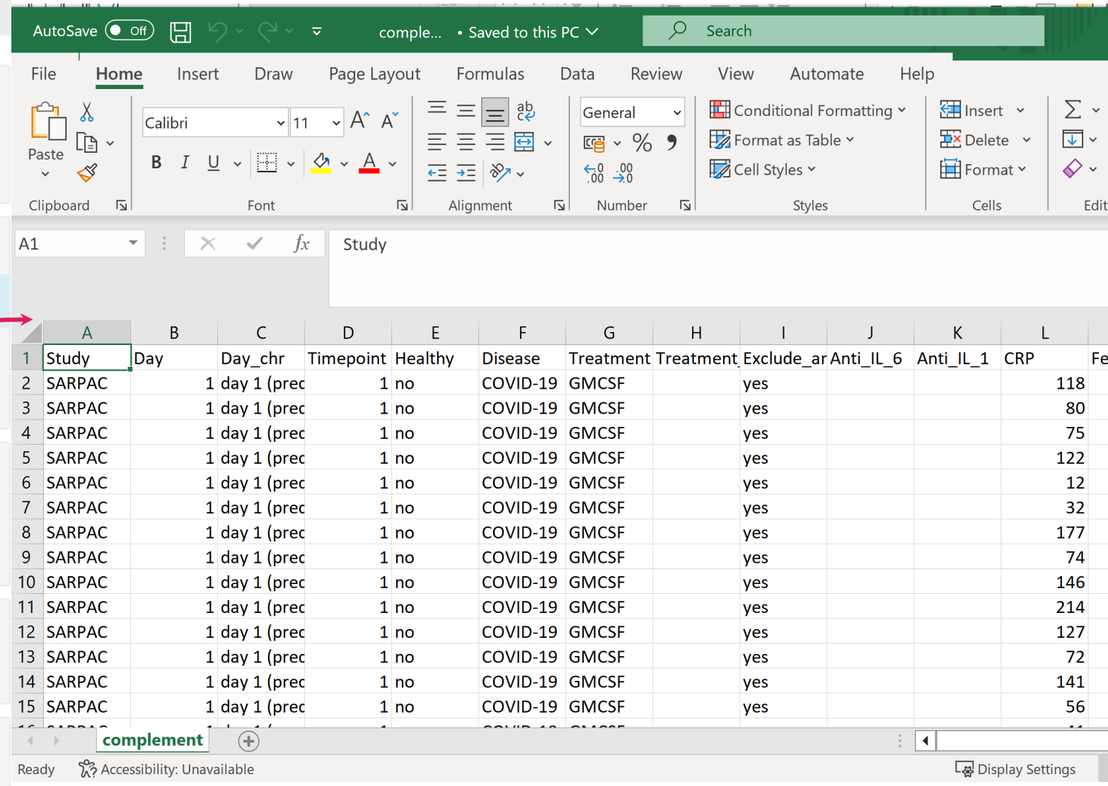


Ghent University Research Data Community

UNIVERSITEIT GENT

DATA REPOSITORY EXAMPLE: ZENODO = GENERAL PURPOSE





	A	B	C	D	E	F	G	H	I	J	K	L
1	Study	Day	Day_chr	Timepoint	Healthy	Disease	Treatment	Treatment_	Exclude_ar	Anti_IL_6	Anti_IL_1	CRP
2	SARPAC	1	day 1 (prec	1	no	COVID-19	GMCSF		yes			118
3	SARPAC	1	day 1 (prec	1	no	COVID-19	GMCSF		yes			80
4	SARPAC	1	day 1 (prec	1	no	COVID-19	GMCSF		yes			75
5	SARPAC	1	day 1 (prec	1	no	COVID-19	GMCSF		yes			122
6	SARPAC	1	day 1 (prec	1	no	COVID-19	GMCSF		yes			12
7	SARPAC	1	day 1 (prec	1	no	COVID-19	GMCSF		yes			32
8	SARPAC	1	day 1 (prec	1	no	COVID-19	GMCSF		yes			177
9	SARPAC	1	day 1 (prec	1	no	COVID-19	GMCSF		yes			74
10	SARPAC	1	day 1 (prec	1	no	COVID-19	GMCSF		yes			146
11	SARPAC	1	day 1 (prec	1	no	COVID-19	GMCSF		yes			214
12	SARPAC	1	day 1 (prec	1	no	COVID-19	GMCSF		yes			127
13	SARPAC	1	day 1 (prec	1	no	COVID-19	GMCSF		yes			72
14	SARPAC	1	day 1 (prec	1	no	COVID-19	GMCSF		yes			141
15	SARPAC	1	day 1 (prec	1	no	COVID-19	GMCSF		yes			56

complement

https://zenodo.org/doi/10.5281/zenodo.8192091

Article: DOI: 10.1126/scitranslmed.adi0252

UNIVERSITEIT GENT

👉 More on how to use data repositories

A wizard can be of help

EMBL-EBI



EMBL-EBI home > Services > Data submission

Data submission

Use this data submission wizard to find the right archive for your data in a few simple steps.

1 What **type of data** do you have?

DNA/RNA sequence | Expression data | Protein data | Structures | Systems

Chemical biology | Ontologies | Images | Multi-omics or other cross-domain study

Other biological research data

Why submit data to an archive?

- Submission of primary data and derived information to public data repositories is an essential step in the scientific process.
- Through submission, the scientific community is fed the raw materials for the building and maintenance of the complete and up-to-date data sets that support searches and analysis on the latest sequences, structures and molecular profiles of living systems.
- Serving as a complement to the literature publication process and supporting early data sharing, the EMBL-EBI offers a number of submission services appropriate for different types and scales of data.

Need help?

If you need help with your data submission, please contact support.

https://www.ebi.ac.uk/submission/

All EMBL-EBI data repositories

Array Express: functional genomics data

BioImage Archive: bioimaging data

BioModels: computational models

BioSamples: reference sample data

BioStudies: biological research data

ChEBI: chemical entities

DGVa: structural genetic variation data

EFO: experimental variables

EGA: human data that requires controlled access

EMPIAR: raw image data

ENA: nucleotide sequence data

EVA: genetic variation data

GO: Gene ontology annotations

GWAS Catalog: Genome-wide association study data

IntAct: molecular interactions

IntEnz: enzyme nomenclature

MetaboLights: metabolomics data

Metagenomics: raw sequence data & associated meta-data

OneDep: electron microscopy, X-ray crystallography & NMR data

PRIDE: protein & peptide identification data

UniProt SPIN: protein sequences & annotations

UniProt: updates or corrections

UNIVERSITEIT GENT

The ELIXIR recommended deposition databases

ELIXIR Deposition Database list

Deposition Database	Data type	International collaboration framework [1]
ArrayExpress	Functional genomics data. Stores data from high-throughput functional genomics experiments.	
BioModels	Computational models of biological processes.	
BioSamples	BioSamples stores and supplies descriptions and metadata about biological samples used in research and development by academia and industry.	NCBI BioSamples database
BioStudies	Descriptions of biological studies, links to data from these studies in other databases, as well as data that do not fit in the structured archives.	
EGA	Personally identifiable genetic and phenotypic data resulting from biomedical research projects.	European Bioinformatics Institute and the Centre for Genomic Regulation
EMDB	The Electron Microscopy Data Bank is a public repository for electron microscopy density maps of macromolecular complexes and subcellular structures.	
ENA	Nucleotide sequence information, covering raw sequencing data, contextual data, sequence assembly information and functional and taxonomic annotation.	International Nucleotide Sequence Database Collaboration
IntAct	IntAct provides a freely available, open source database system and analysis tools for molecular interaction data.	The International Molecular Exchange Consortium
MetaboLights	Metabolite structures and their reference spectra as well as their biological roles, locations and concentrations, and experimental data from metabolic experiments.	
PDBe	Biological macromolecular structures.	wwPDB
PRIDE	Mass spectrometry-based proteomics data, including peptide and protein expression information (identifications and quantification values) and the supporting mass spectra evidence.	The ProteomeXchange Consortium

Further information: Gavin Farrell (gavin.farrell@elixir-europe.org)

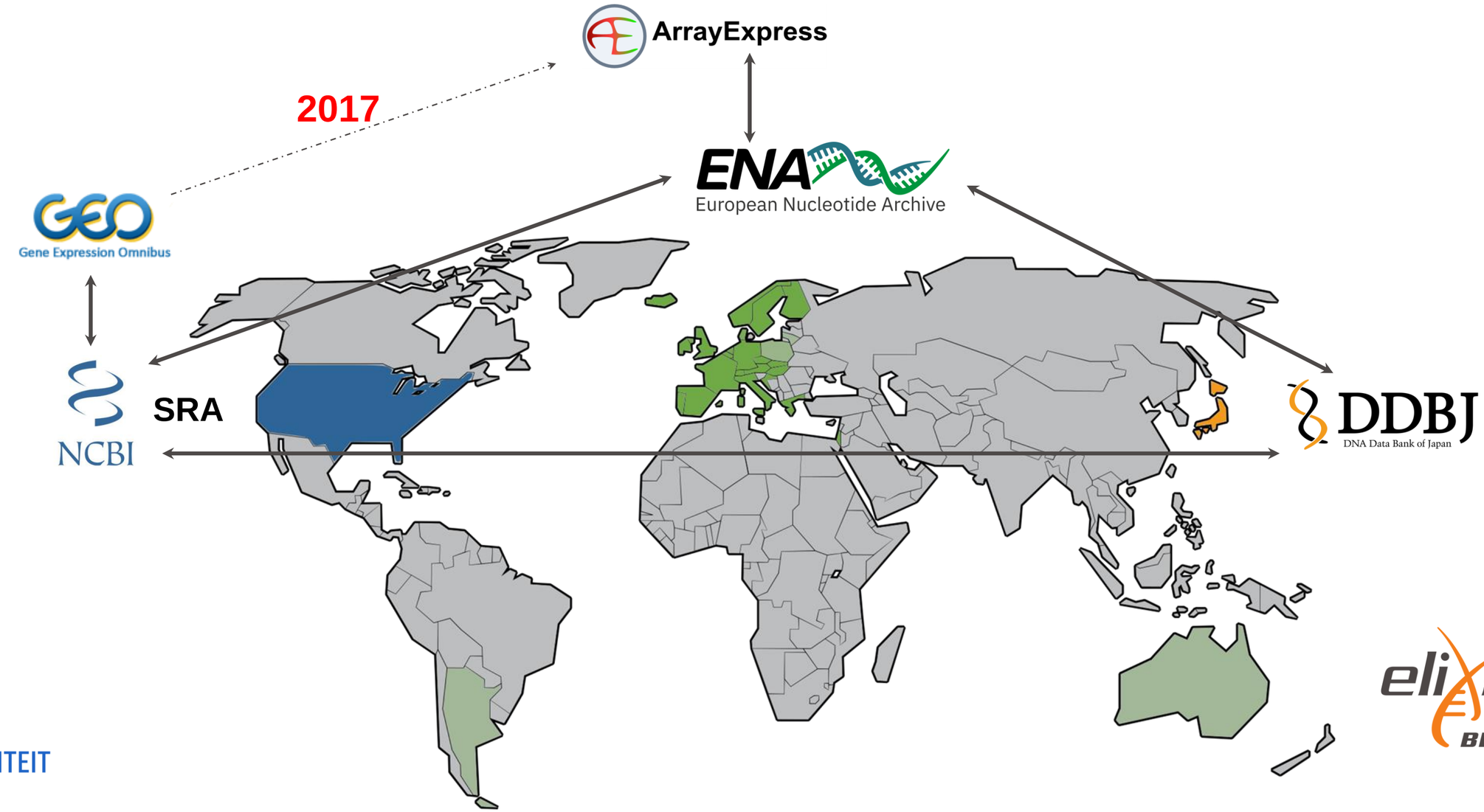
[1] An International collaboration framework enables content sharing on a formal level. This is often signified by a shared Accession Number system, such that data deposited in one database becomes part of the shared data collection, and is also available through other partner portals.

https://elixir-europe.org/platforms/data/elixir-deposition-databases

Focus on experimental data

elixir BELGIUM

Multiple repositories with a similar data type - Genomics



UNIVERSITEIT GENT

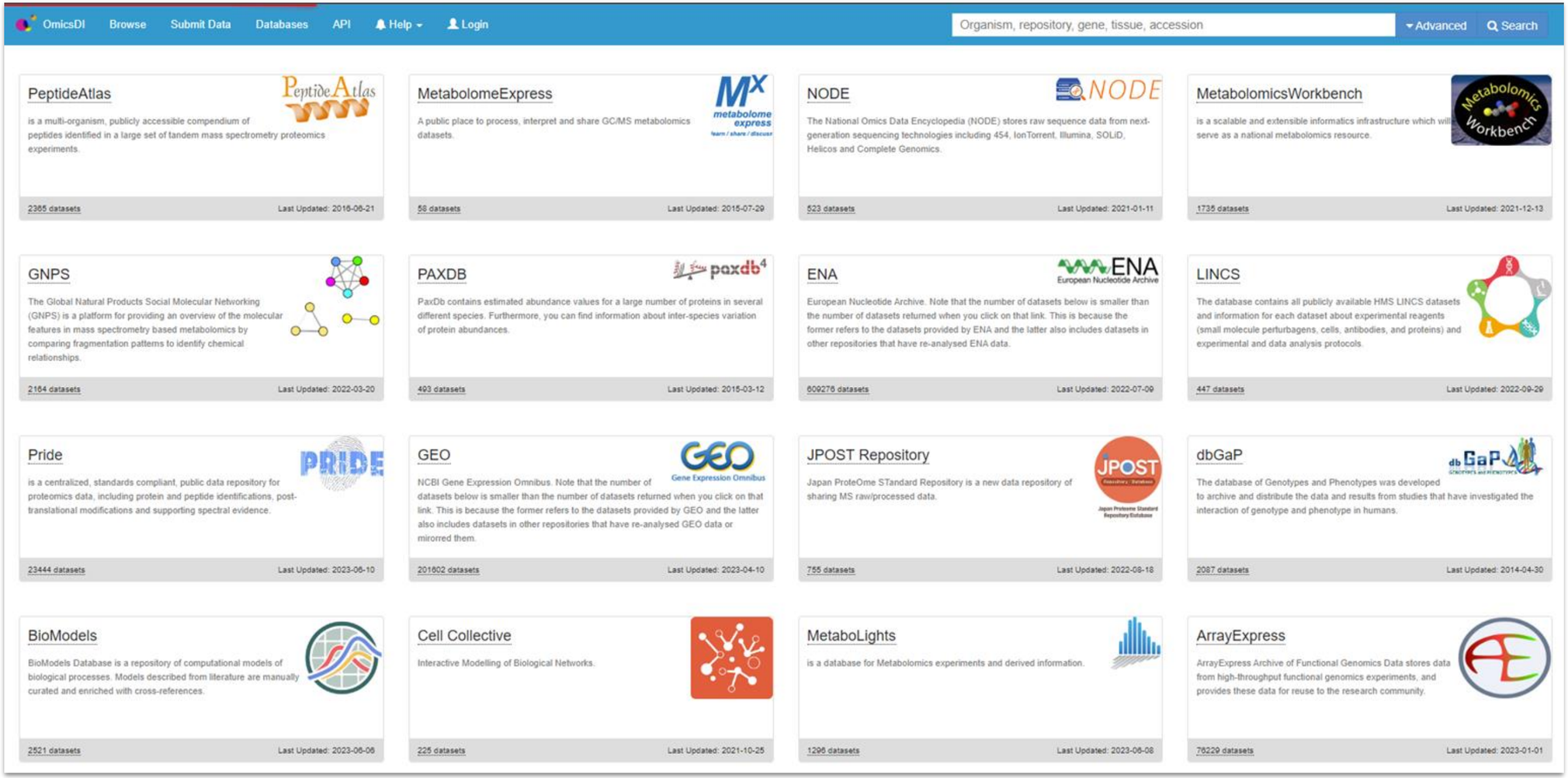
elixir BELGIUM

Thank you to Alexander Botzki for original slide material

Multiple repositories with a similar data type - multi-omics

The Omics Discovery Index - OmicsDI

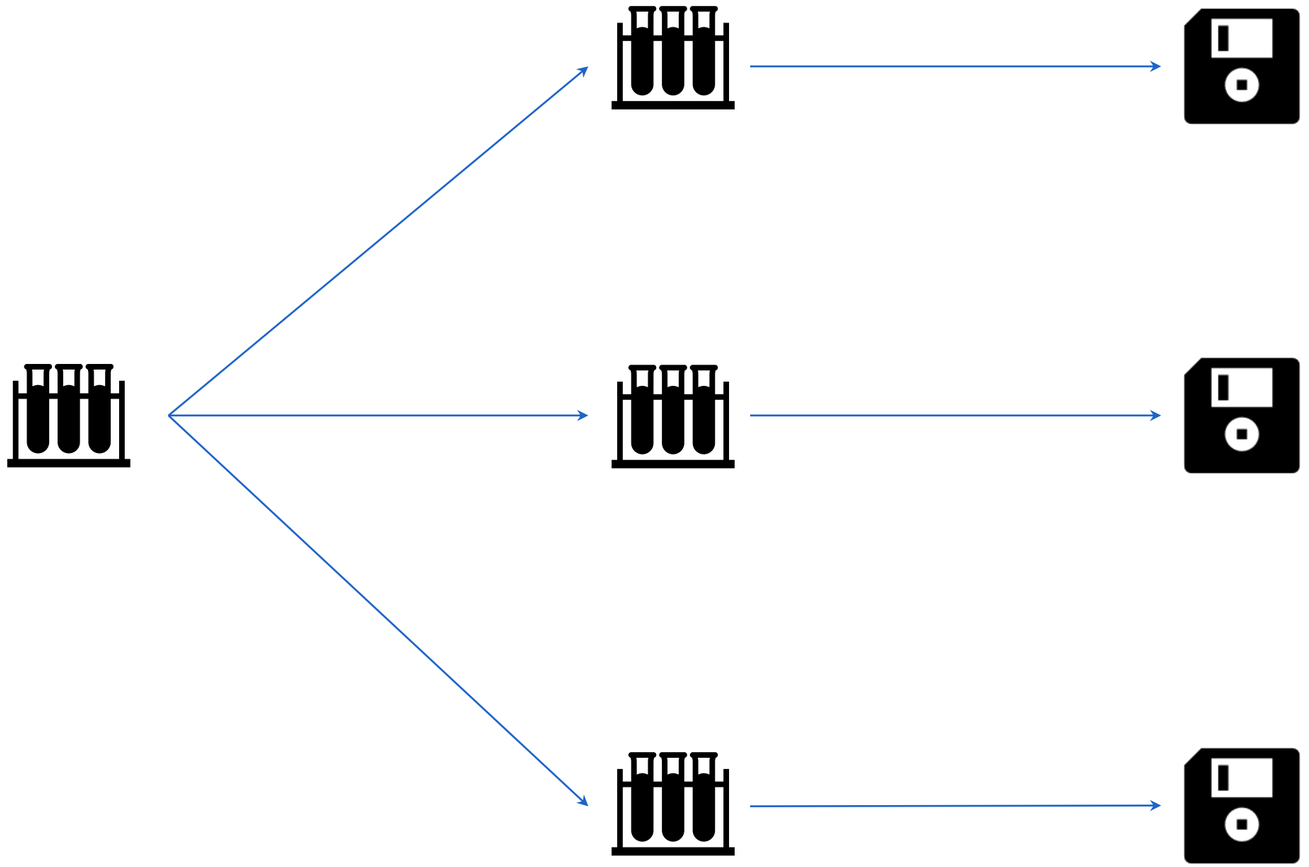
25 Resources
>3M Datasets
4 Continents



UNIVERSITEIT GENT

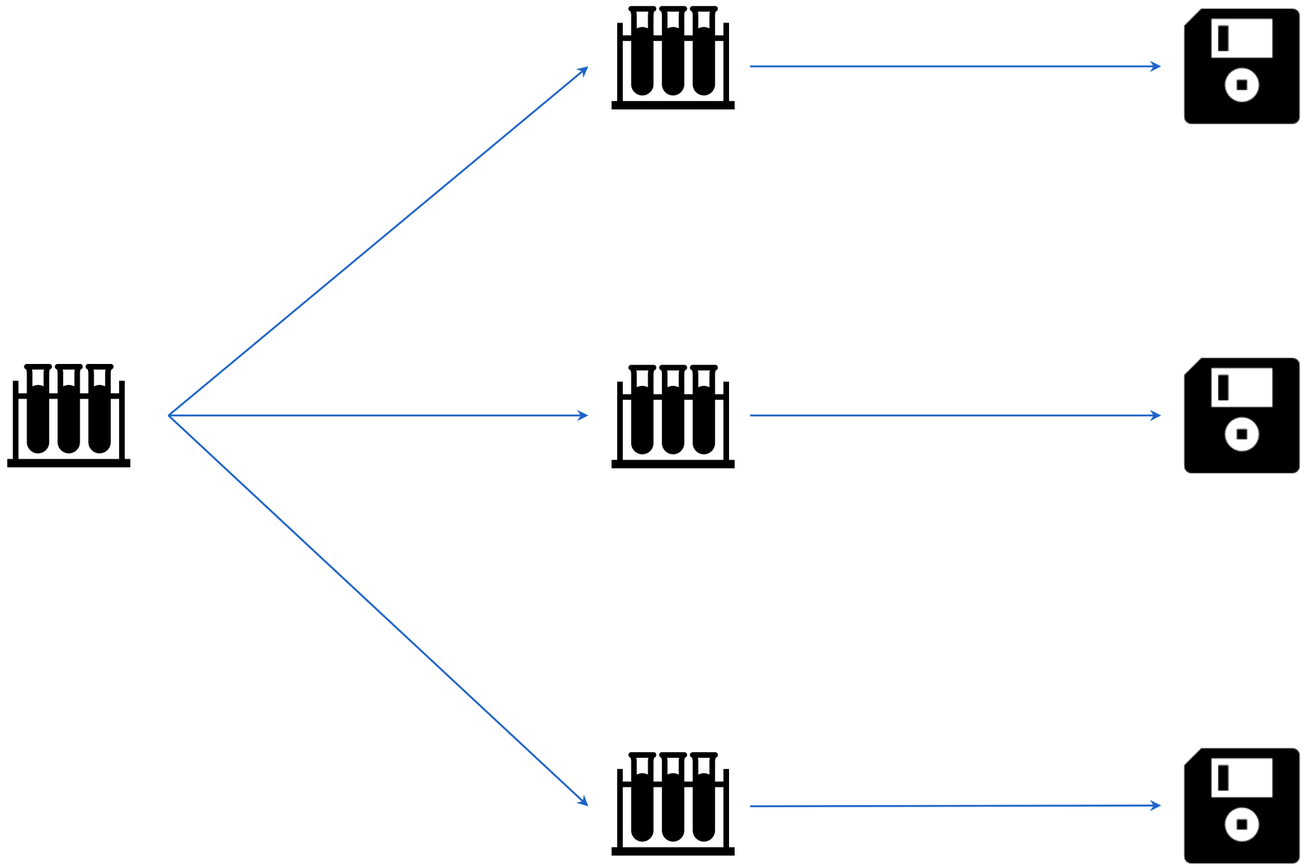
https://www.omicsdi.org/

Multiple datatypes in 1 study



UNIVERSITEIT
GENT

Multiple datatypes in 1 study



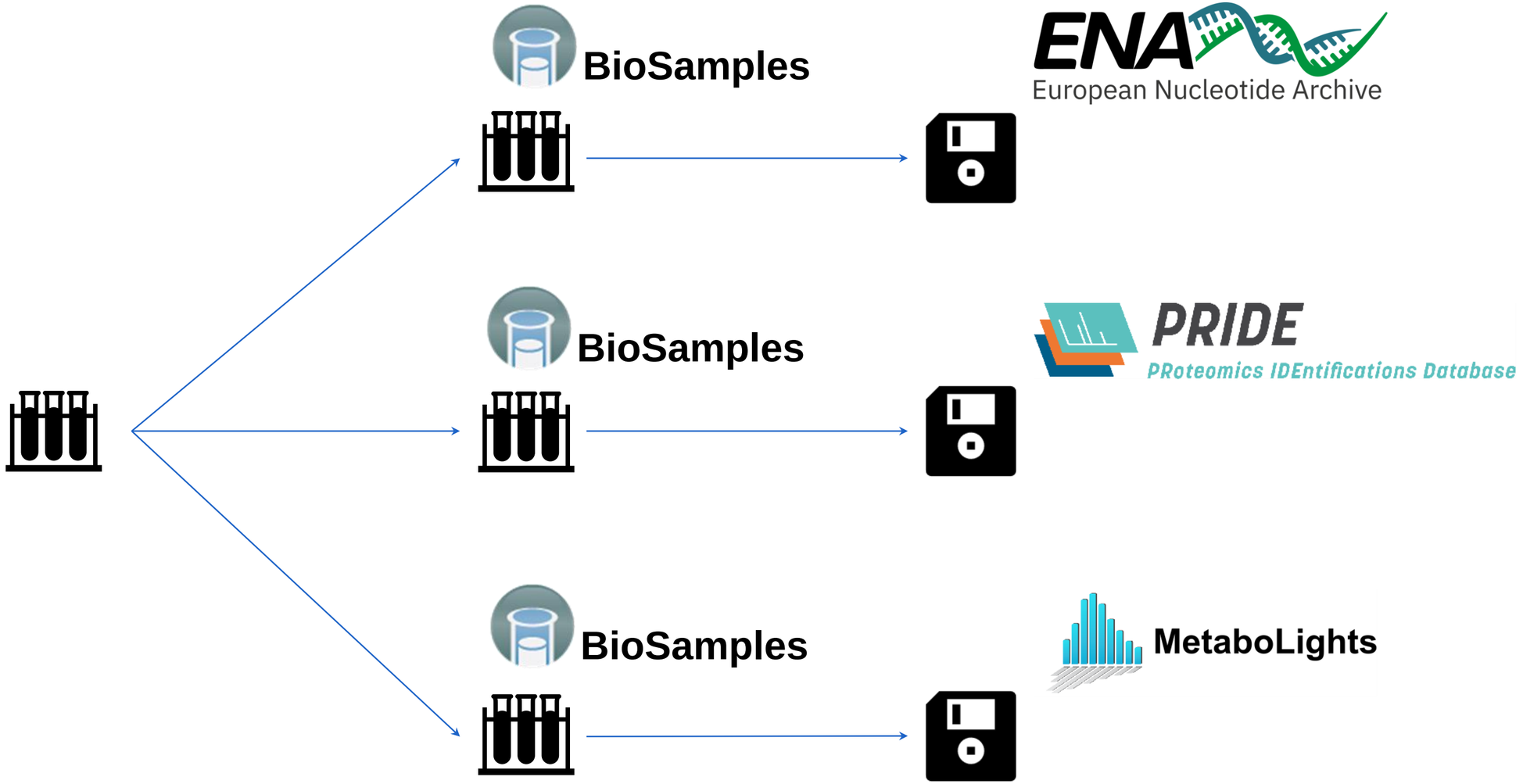
ENA
European Nucleotide Archive

PRIDE
PRoteomics IDEntifications Database

MetaboLights

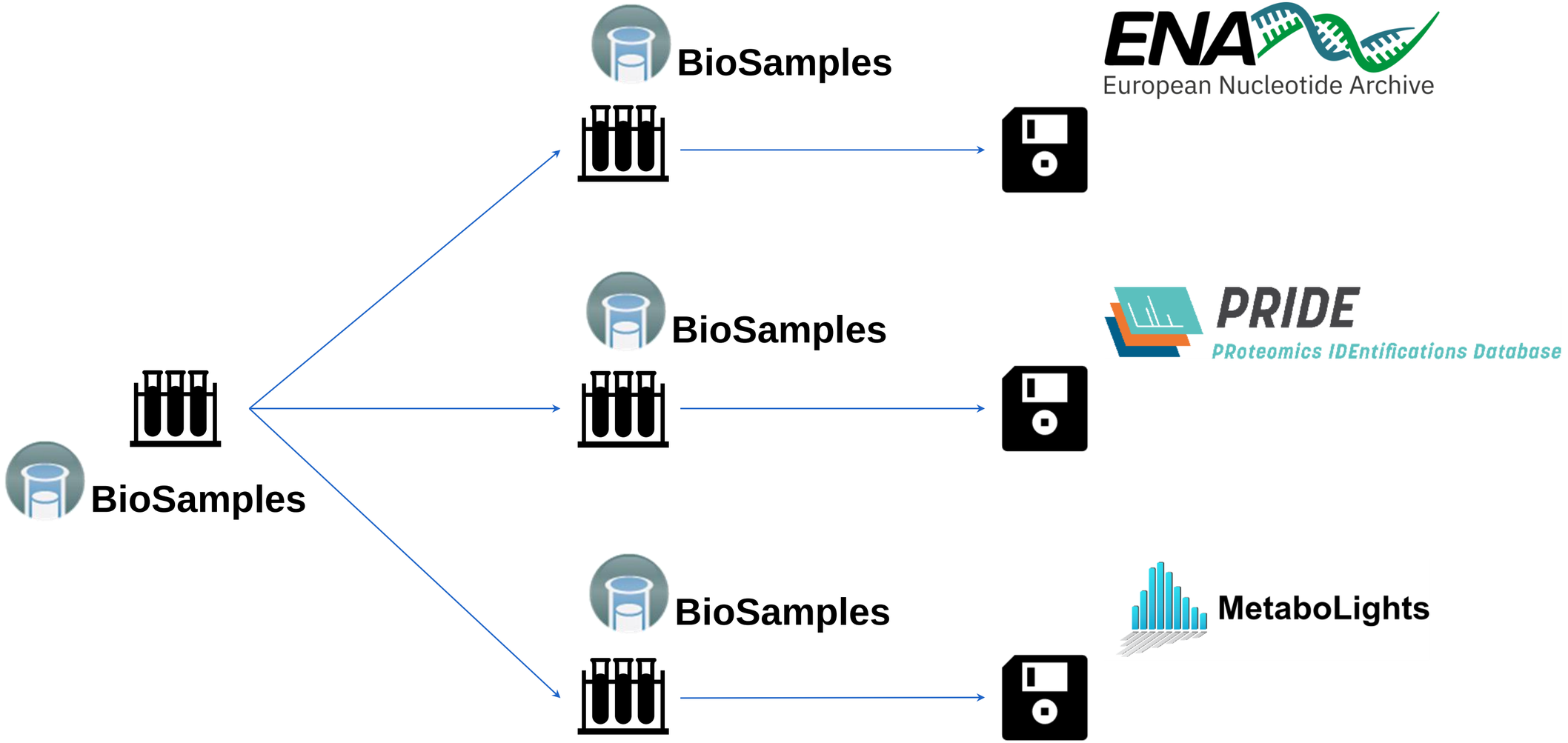
UNIVERSITEIT
GENT

Multiple datatypes in 1 study



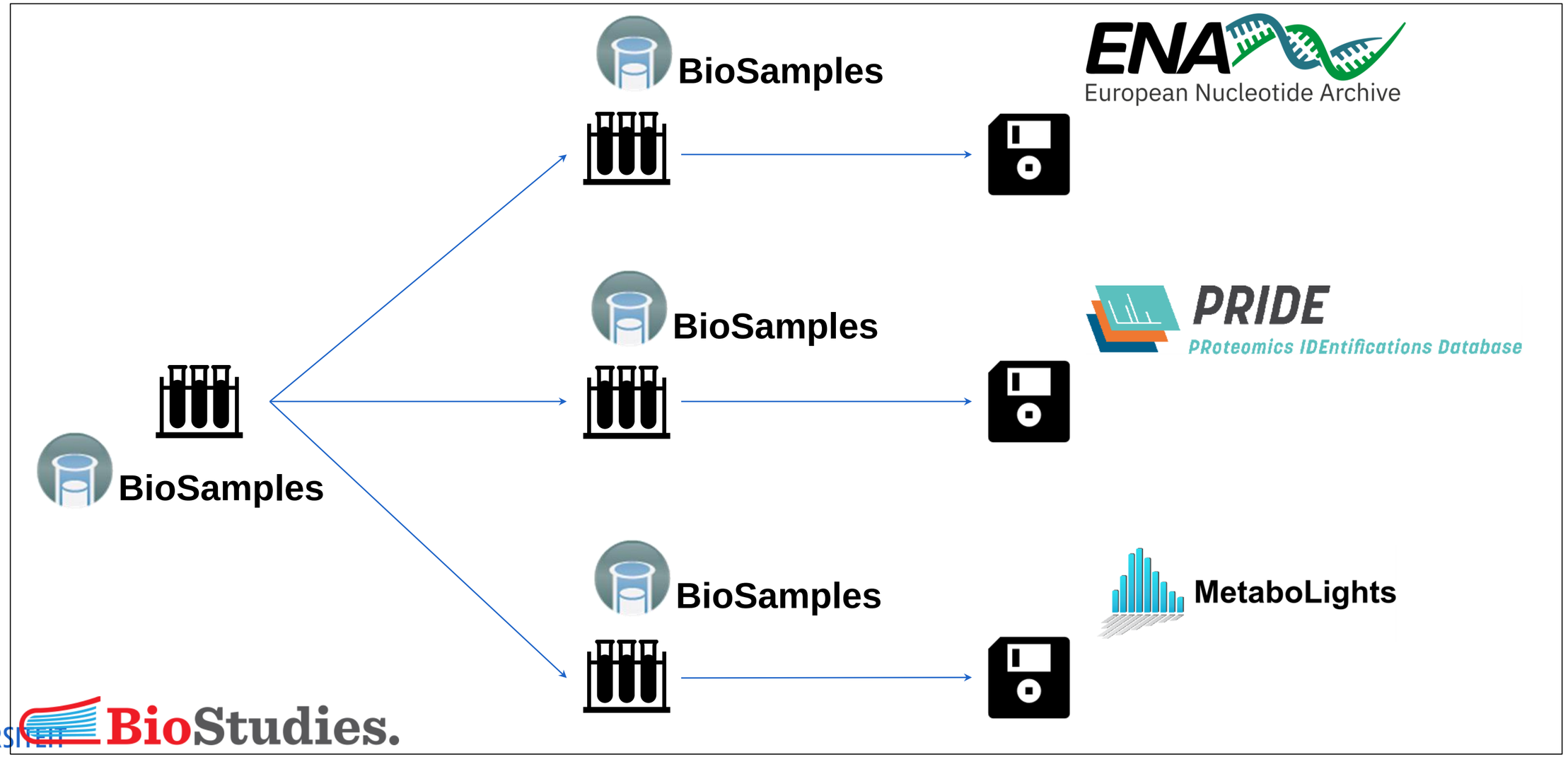
UNIVERSITEIT GENT

Multiple datatypes in 1 study



UNIVERSITEIT GENT

Multiple datatypes in 1 study



UNIVERSITEIT GENT

BioStudies.

TERMS FOR REUSE

UNIVERSITEIT
GENT

LICENSES

- Allow an author to give other people the right to **share, use, and build** upon a work that they have created.
- Define what can and cannot be done with data in a clear way.
- Licenses are **not automatically assigned**. The owner of rights on protected dataset must attach a license.

- Data ‘publicly available’ without license cannot be accessed, used and shared under copyright or database laws. Licenses allow to **expand rather than restrict** use.

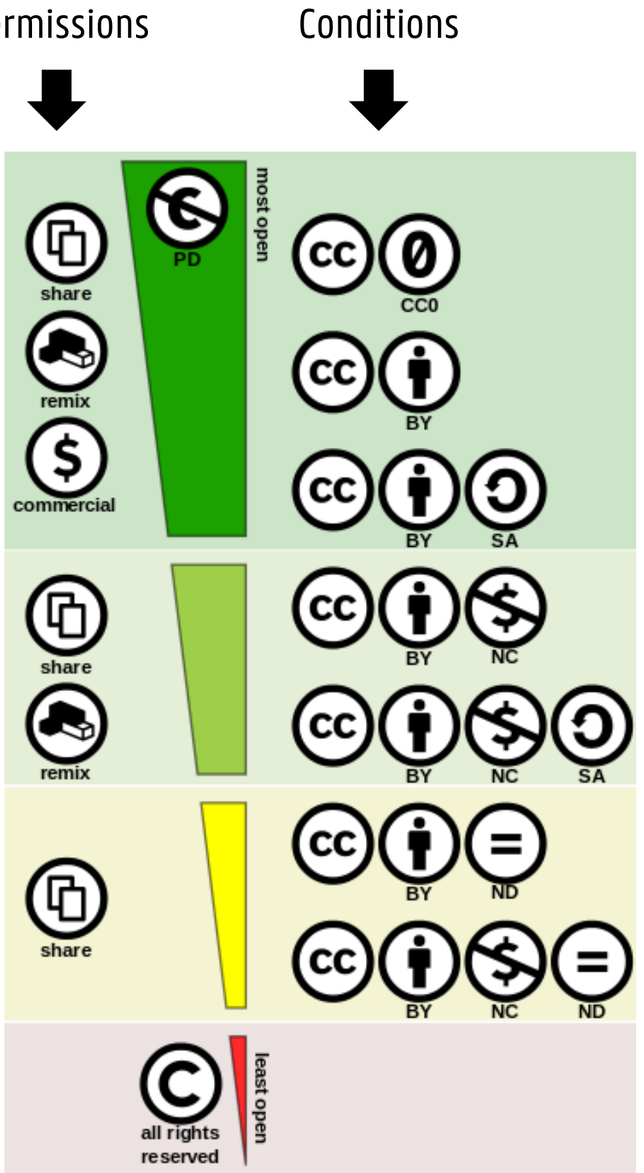
OPEN LICENSING: CC BY, CC 0 OR EQUIVALENT

Creative Commons (CC) Licenses = standard and commonly used to grant permissions under certain conditions for data and other research output:

- **CC 0: "No Rights Reserved":** to place data as completely as possible in the public domain, so that others may freely build upon, enhance and reuse the works for any purposes without restriction under copyright or database law.

- **CC-BY: Attribution (BY)**: give proper credit to the original creator

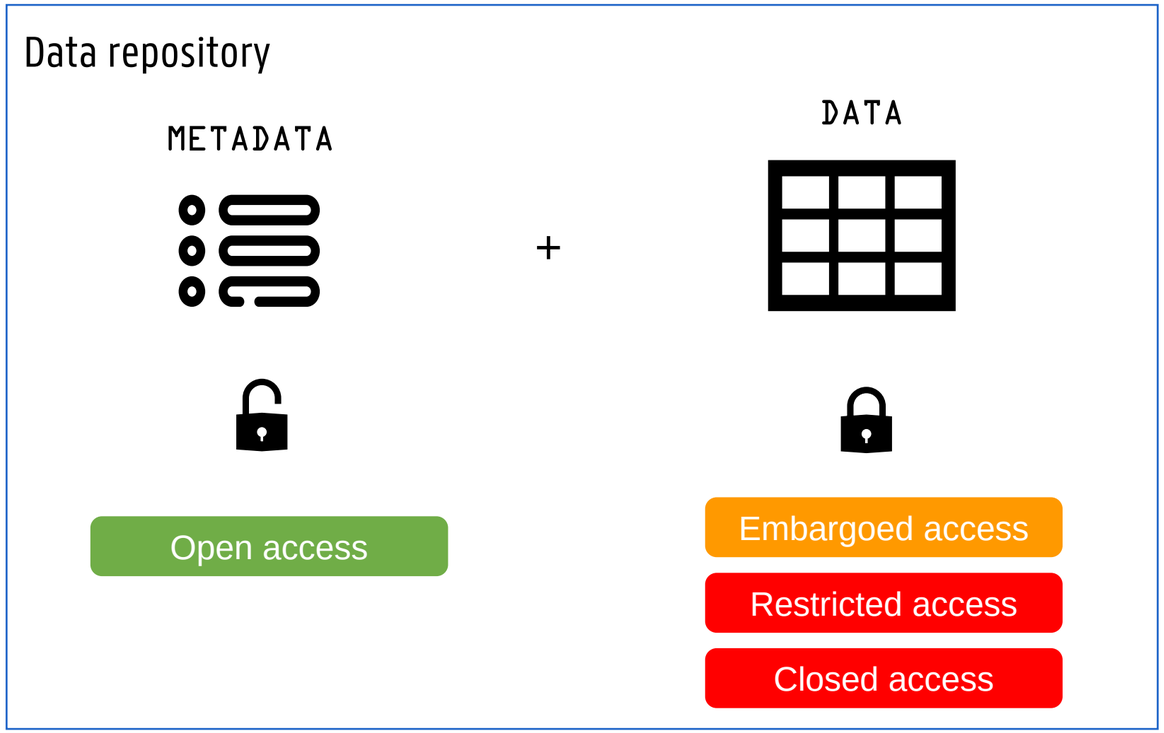
Note: Licenses are **not automatic.** No license = no reuse allowed



UNIVERSITEIT GENT

METADATA OPEN UNDER CC 0 OR EQUIVALENT

- to the extent legitimate interests or constraints are safeguarded
- provide information about the licensing terms, amongst others



UNIVERSITEIT GENT

DIFFERENT TYPES OF LICENSES

- **Standard licenses**: terms and conditions are already defined.
  - Simplicity and transparency.
  - Machine readability and interoperability
  - Most widely known standard **licenses for data** are **Creative Commons (CC)** and Open Data Commons (ODC).
- **Custom-made** licenses: can be very restrictive
  - E.g. Data transfer agreement (DTA)
- Other licenses:
    - Open software licenses (e.g. GNU General Public License, Apache).
    - Open Government licenses. E.g. UK Open Government license.

UNDERSTANDING THE DATA

UNIVERSITEIT
GENT

PROVIDE **INFORMATION** VIA THE REPOSITORY ABOUT ANY RESEARCH OUTPUT/TOOLS/INSTRUMENTS NEEDED TO RE-USE OR VALIDATE THE DATA

Towards reproducible research



The Turing Way project illustration by Scriberia. Used under a CC-BY 4.0 licence.
DOI: 10.5281/zenodo.3332807.

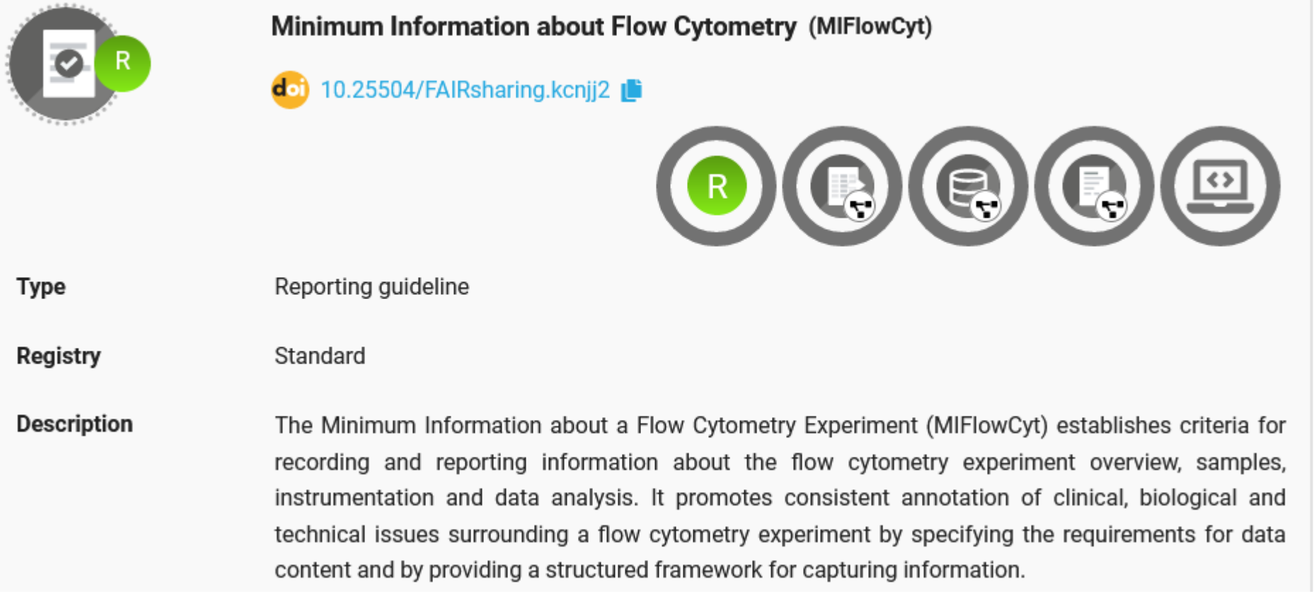
Information should include a detailed description of the research output/tool/instrument (data, software, algorithms, protocols, models, workflows, electronic notebooks and others), how to access it, any dependencies on commercial products, potential version/type, potential parameters etc.

UNIVERSITEIT GENT

MINIMAL INFORMATION ABOUT …

- Flow Cytometry
- Sequencing Experiment
- …

https://fairsharing.org/search?q=minimal%20information%20about



Minimum Information about Flow Cytometry (MIFlowCyt)

10.25504/FAIRsharing.kcnjj2

Type	Reporting guideline
Registry	Standard
Description	The Minimum Information about a Flow Cytometry Experiment (MIFlowCyt) establishes criteria for recording and reporting information about the flow cytometry experiment overview, samples, instrumentation and data analysis. It promotes consistent annotation of clinical, biological and technical issues surrounding a flow cytometry experiment by specifying the requirements for data content and by providing a structured framework for capturing information.



Minimal Information about a high throughput SEQuencing Experiment (MINSEQE)

10.25504/FAIRsharing.a55z32

Type	Reporting guideline
Registry	Standard
Description	MINSEQE describes the Minimum Information about a high-throughput nucleotide SEQuencing Experiment that is needed to enable the unambiguous interpretation and facilitate reproduction of the results of the experiment. By analogy to the MIAME guidelines for microarray experiments, adherence to the MINSEQE guidelines will improve integration of multiple experiments across different modalities, thereby maximising the value of high-throughput research. Metadata collection template includes critical information such as: 1) The description of the biological system, samples

UNIVERSITEIT GENT

DOCUMENTATION VS METADATA

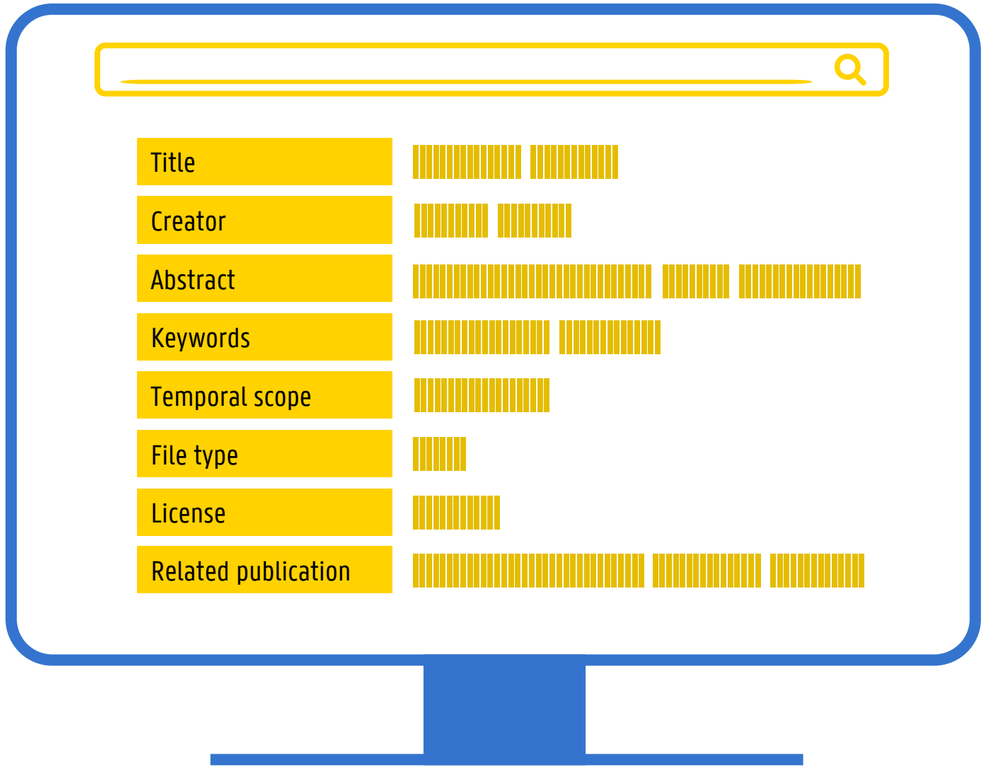
DOCUMENTATION

Any contextual information that allows data to be understandable and reusable.



METADATA

Information about data, recorded in a highly structured way, using a set of defined information fields or elements to be machine readable.

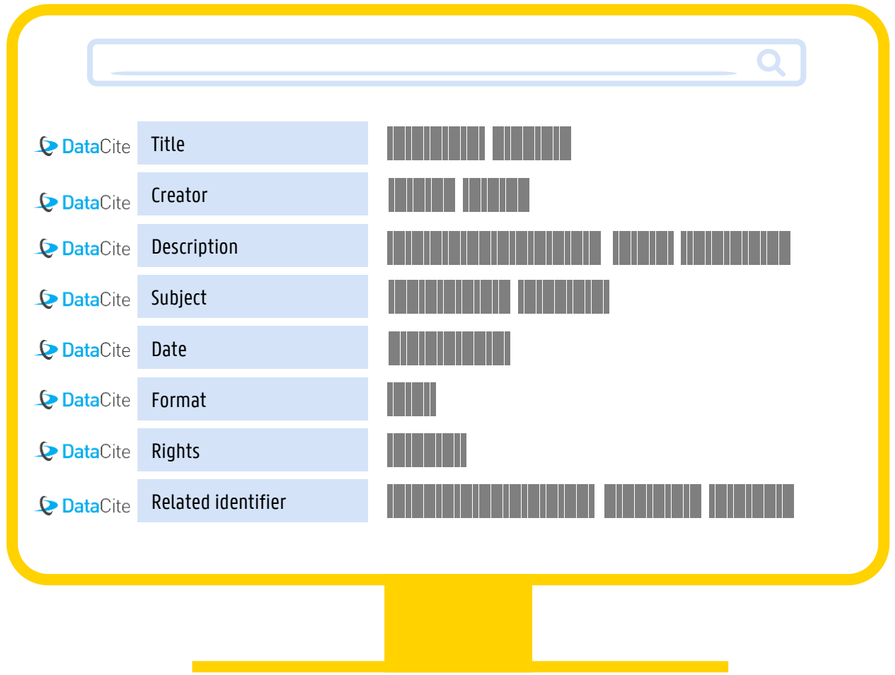


UNIVERSITEIT GENT

METADATA STANDARDS FOR DATASETS

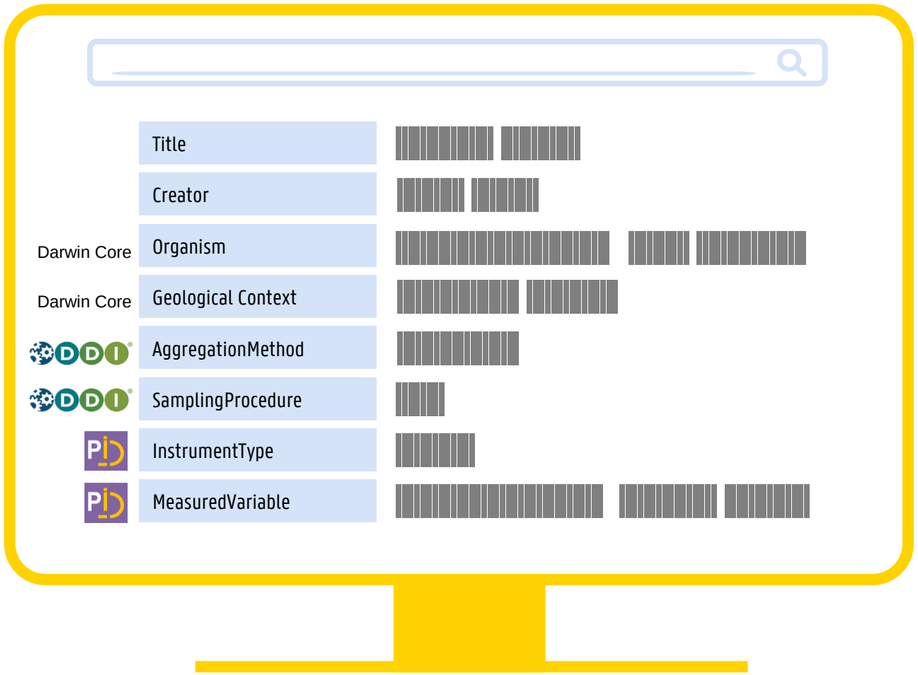
Domain-agnostic standards:

DataCite, Dublin Core, schema.org, etc.



Domain-specific standards:

Darwin Core, Data Documentation Initiative (DDI), Ecological Metadata Language, Persistent Identification of Instruments (PIDINST), etc.

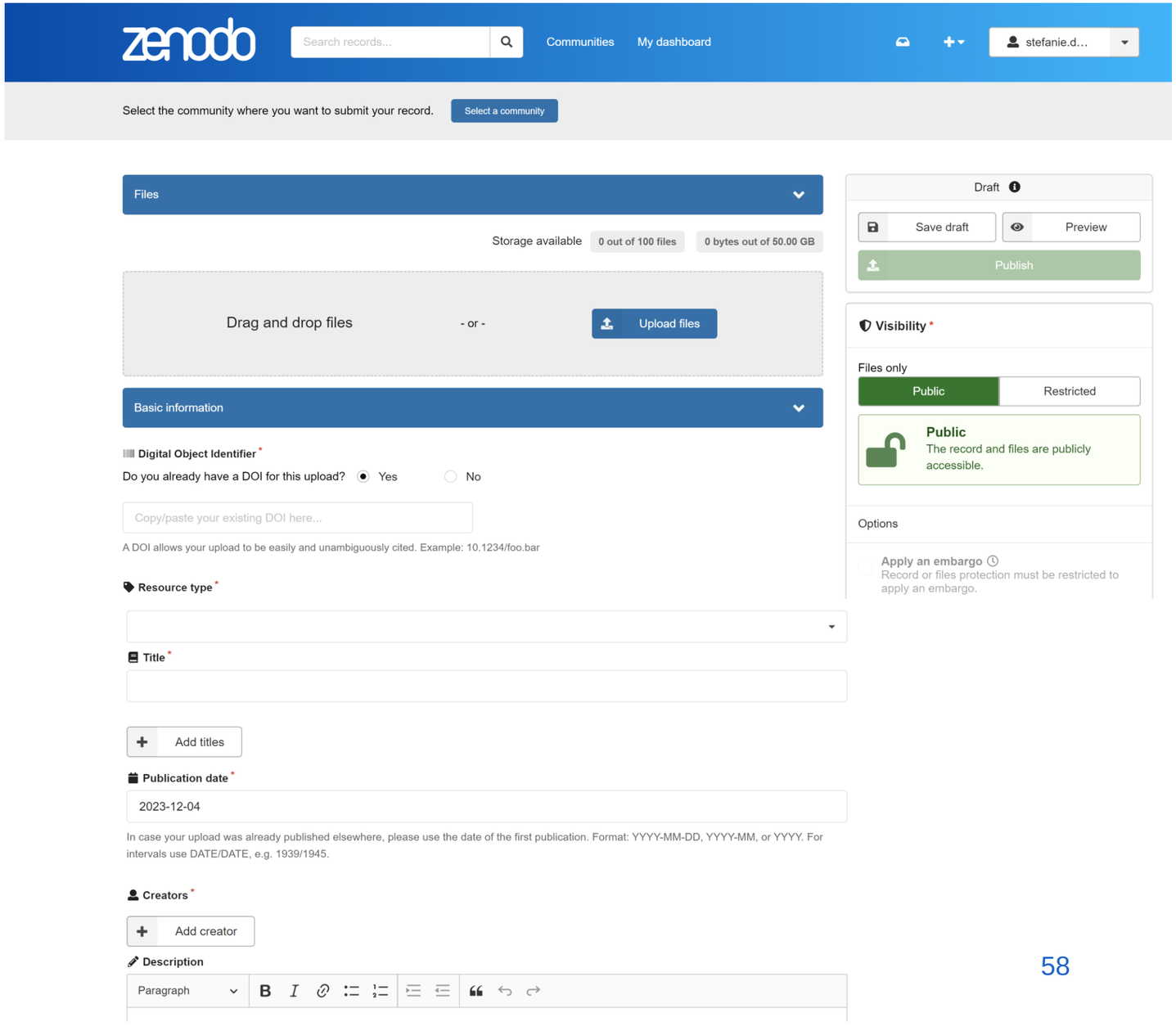


UNIVERSITEIT GENT

METADATA IN ZENODO – A GENERIC DATA REPOSITORY

- Repositories: infrastructure to allow us to share & find data
- Provide **services to create & manage metadata**
- Typically, by filling a form with
  - Pre-defined fields
  - Lists of controlled terms to choose from (e.g. standard licenses)
- Metadata adheres to a specific metadata standard (interoperability)

Zenodo data upload form

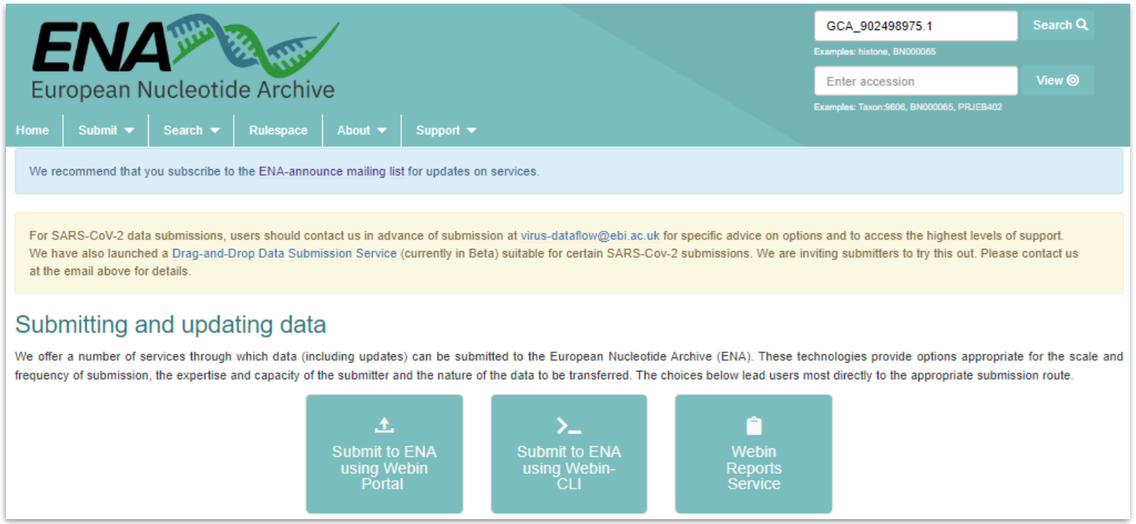


UNIVERSITEIT GENT

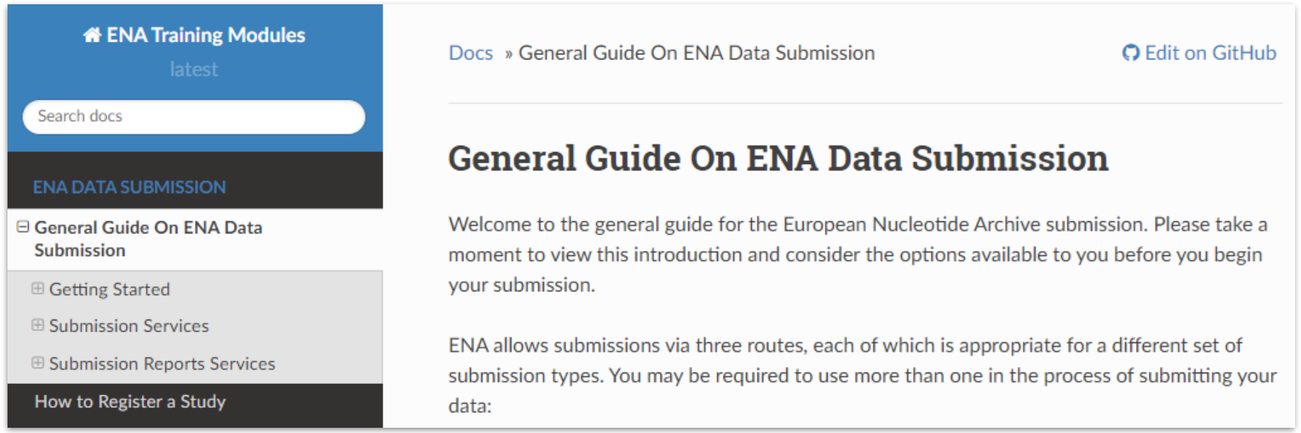
Data submission

Different repositories = different routes

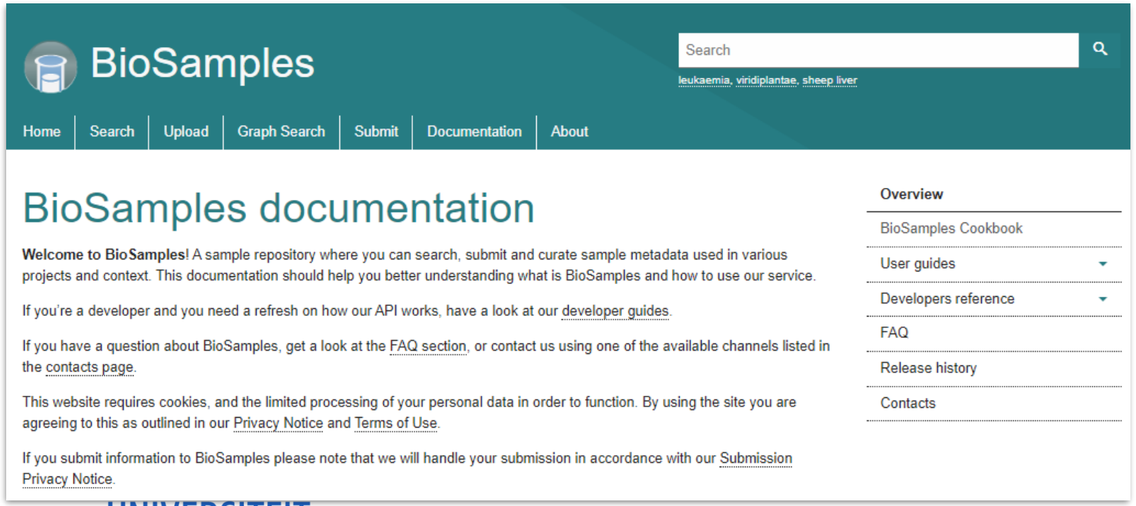
- What file formats are expected?
- What metadata is expected?
- Can ontologies and controlled vocabularies be used?
- Are there costs involved?
- Is there a license to be applied? Is it implicit?



https://www.ebi.ac.uk/ena/browser/submit



https://ena-docs.readthedocs.io/en/latest/submit/general-guide.html



https://www.ebi.ac.uk/biosamples/docs

UNIVERSITEIT GENT

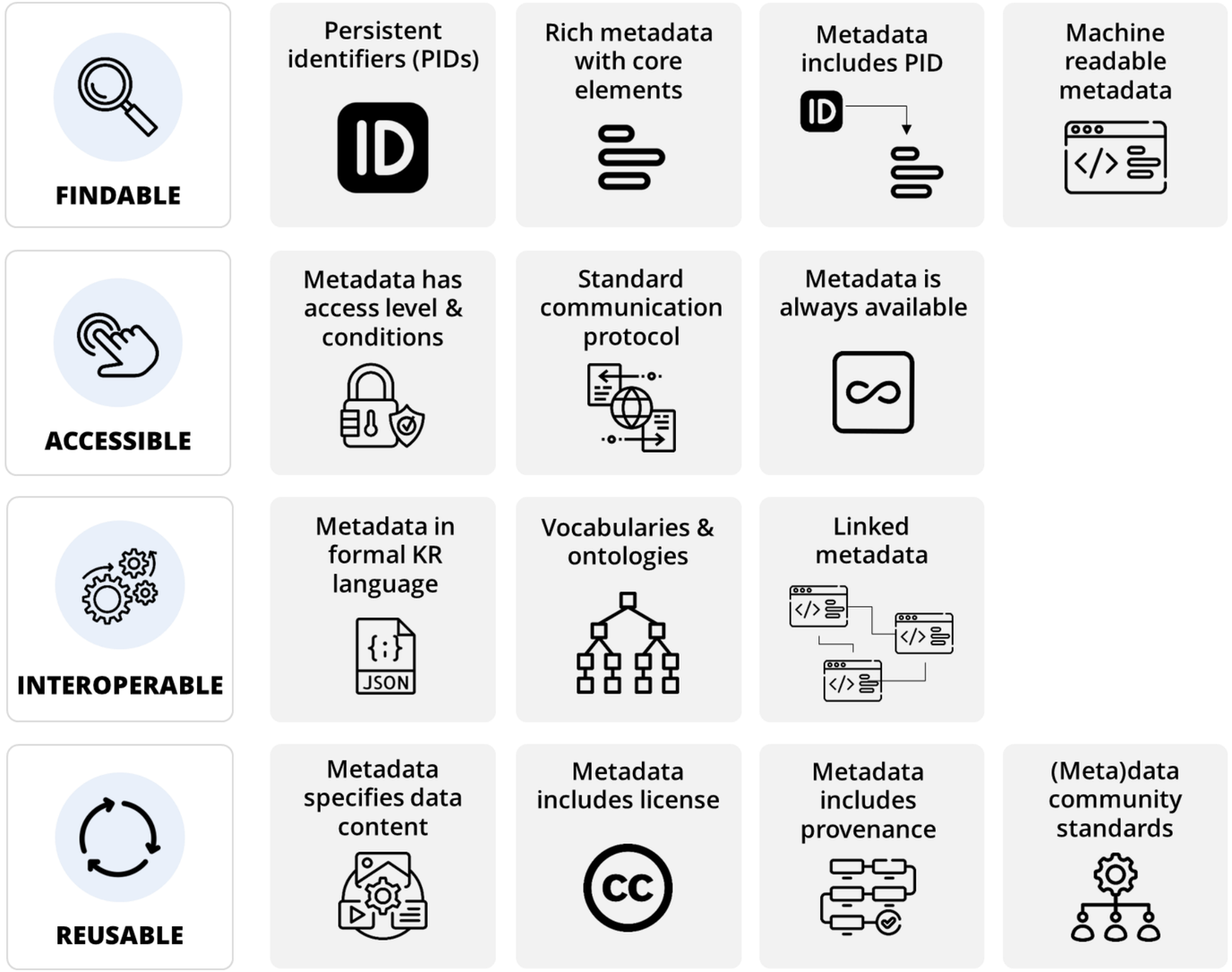
elixir BELGIUM

METADATA: CHECK REPOSITORY SUBMISSION GUIDELINES

Example for nucleotide variation deposition: **EVA submissions**

- Prepare valid VCF file(s), which can be validated prior to submission using the EVA VCF validation suite.
- Complete a metadata template describing the samples and analyses in your study. Please provide as much metadata as possible since this information is extremely useful for downstream analysis and is directly related to the frequency at which datasets archived at EVA are reused. For reference, here is an **example** of a completed metadata template.

FAIR DATA PRINCIPLES



UNIVERSITEIT GENT

PREPARING DATA FOR DEPOSIT IN A REPOSITORY - CHECKLIST

- Gather all relevant (meta)data needed for reproducibility:
  - experimental method details, raw data files, organized data tables, scripts and code, data visualizations, and statistical output
- Migrate data to recommended, reusable formats
- Organize files in a logical schema
- Provide a separate readme file



Jørgen Stamp - Digitalbevaring.dk

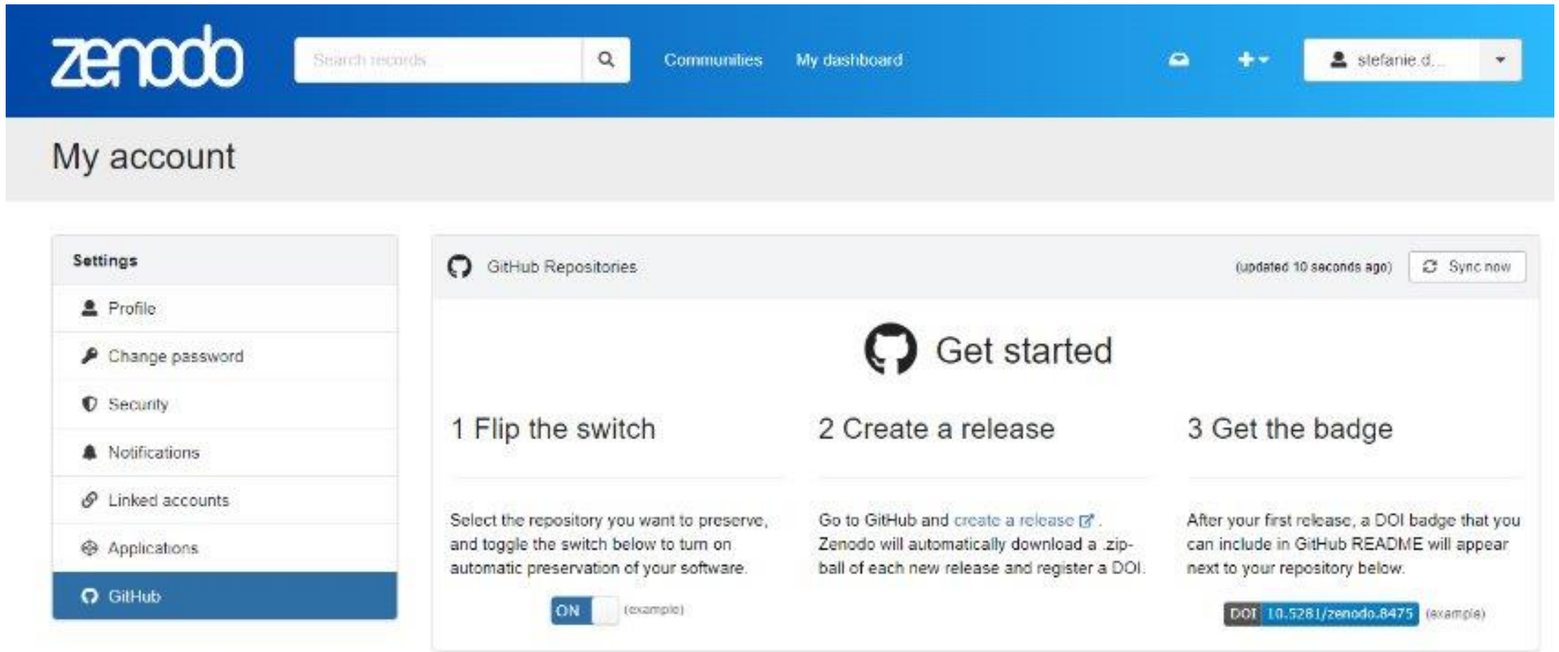
More info at: https://datadryad.org/stash/best_practices

UNIVERSITEIT GENT

SHARING SOFTWARE

GitHub – Zenodo integration: https://guides.github.com/activities/citable-code/

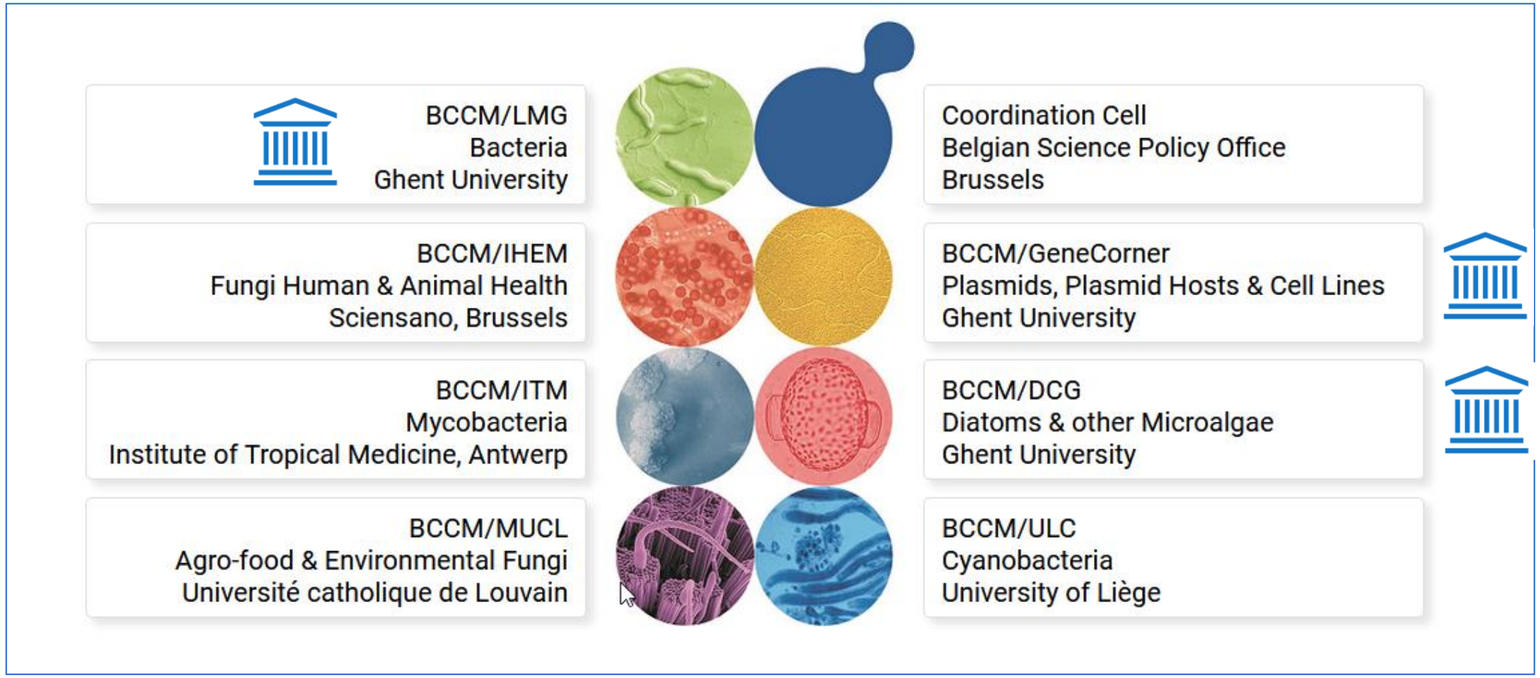
GitHub +



UNIVERSITEIT GENT

SHARING BIOLOGICAL RESOURCES

https://bccm.belspo.be/



UNIVERSITEIT GENT

BCCM

EXAMPLES

EXAMPLE 1

- https://www.sciencedirect.com/science/article/pii/S0092867421014811?via%3Dihub#app2
- Table S1. Differentially expressed genes (DEGs) and proteins (DEPs) per cell type in total mouse and human liver cell atlases, related to Figures 1, 4, S2, and S6.
- Table S2. Differentially expressed genes (DEGs) and proteins (DEPs) per cell type in mouse and human myeloid cell atlases, related to Figures 2, 4, 5, S3, S6, and S8.
- … additional resources:
- The sc/snRNA-sequencing, CITE-seq FCS files and spatial transcriptomics datasets will be made available for visualization, analysis and download at www.livercellatlas.org.

UNIVERSITEIT GENT

EXAMPLES 2: ZENODO

- Example restricted access on Zenodo: https://zenodo.org/record/6697642
- Example open data on zenodo: https://zenodo.org/record/6138081
-
- Immunoblot test results: https://zenodo.org/record/3995090

UNIVERSITEIT GENT

EXAMPLE 3

- Article Science Translational Medicine: https://www.science.org/doi/10.1126/scitranslmed.adi0252

- https://zenodo.org/record/8192092 datasets plus readme
- https://www.uzgent.be/nieuws/eigen-afweer-veroorzaakt-longschade-bij-covid-patienten nieuwsartikel UZ gent
  - Terwijl traditioneel een publicatie voldoende is, hebben de onderzoekers **alle data ook publiek beschikbaar** gemaakt via een website. “De meest geavanceerde - en dure - biologische technieken leveren makkelijk duizenden meetpunten op. Deze data liggen ergens begraven in de publicatie als addendum, maar worden vaak niet bekeken, wat zonde is,” legt Karel Van Damme uit.
- https://complementatlas.com/ project website: datasets, verwante informatie, visualisatie van data
- Github: https://github.com/kfvdamme/complementatlas
- Clinical trials
  - Effect of anti-interleukin drugs in patients with COVID-19 and signs of cytokine release syndrome (COV-AID): a factorial, randomised, controlled trial (Declercq et al., 2021).
    - ClinicalTrials.gov NCT04330638 results: https://clinicaltrials.gov/study/NCT04330638?tab=results
    - EudraCT 2020-001500-41 results: https://www.clinicaltrialsregister.eu/ctr-search/trial/2020-001500-41/results
  - Loss of GM-CSF-dependent instruction of alveolar macrophages in COVID-19 provides a rationale for inhaled GM-CSF treatment (SARPAC) (Bosteels et al., 2022).
    - EudraCT: https://www.clinicaltrialsregister.eu/ctr-search/trial/2020-001254-22/results
    - Clinicaltrials.gov NCT04326920 : https://clinicaltrials.gov/study/NCT04326920?term=NCT04326920&rank=1&tab=results
  - Efficacy and safety of the investigational complement C5 inhibitor zilucoplan in patients hospitalized with COVID-19: an open-label randomized controlled trial (ZILUCOV) (De Leeuw et al., 2022).
    - Clinical trials: NCT04382755

UNIVERSITEIT GENT

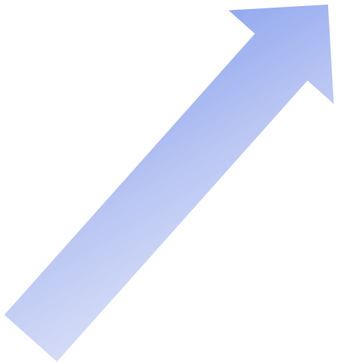
SUMMARY

UNIVERSITEIT
GENT

TAKE HOME MESSAGES

- Plan! Create a DMP, but don't reinvent the wheel
- Be aware of regulatory restraints
- Make use of persistent identifiers
- Deposit data in repositories (as much as possible, or at least the metadata)
- Register in Biblio

Research Impact



UNIVERSITEIT GENT

THE BIG PICTURE
HOW TO PREPARE YOUR DATA FOR POSTING IN A REPOSITORY

Plan for data management, keep your DMP updated

Explore potential data repositories and their requirements & guidelines.

- Inc. use of standards, vocabularies, etc.

Explore regulatory requirements

re3data.org
REGISTRY OF RESEARCH DATA REPOSITORIES

FAIRsharing

DCC

Adopt domain standards during data collection & processing
Implement regulatory requirements

Document as you go

- Notebook
- Codebook, data dictionary
- Workflows
- Protocols
- Readme files
- Data overview with metadata
- File organization and naming

Deposit data & documentation in repository; describe it with rich metadata, incl. access & reuse conditions

Register data in Biblio

UNIVERSITEIT GENT

UGent Data Stewards

RESEARCH DEPARTMENT - UNIVERSITY LIBRARY

rdm.support@ugent.be

@UGentRDM

ugent.be/en/research/datamanagement

www.ugent.be

UNIVERSITEIT
GENT

KNOWLEDGE 2 CONNECT

We are grateful to our sponsors:

Karger

Wolters Kluwer

EBSCO INFORMATION SERVICES

Ovid®

VZW VRIENDEN van de Biomedische Bibliotheek

ELSEVIER

UNIVERSITEIT GENT

GLOSSARY

- **Metadata**: data about data, a structured and machine-readable form of documentation
- **Keywords**: Keywords can be added as metadata in most data repositories. They allow to describe your data more specifically than what is possible through other (generic) metadata fields.
- **Machine-readable**: automatic computer processing can extract information from the metadata ensuring a cross-linking between different research outputs
- **Repository**: a database infrastructure that collects, manages and stores data for preservation and sharing, a generic term to denote such infrastructure, independent of the used technology
- **FAIR**: Findable, Accessible, Interoperable, Reusable
- **Persistent identifier**: a long-lasting and reliable reference to, amongst others, a resource, a person or an institution, providing the information required to reliably identify, verify and locate it.
- **Globally unique identifier**: ensures that there are no two identical identifiers that point to different digital objects
- **DOI: Digital Object Identifier**. A data repository can assign a DOI to the data. Thereby, data is registered and can be uniquely identified for the long term. An identifier can, for instance, be resolved to a landing page (e.g. web page), providing access to metadata, data files, etc., all of which is indexed and searchable.
- **Provenance**: information about the origin, history or workflow that generated the data
- **Metadata harvesting and indexing**: refers to gathering metadata from multiple places or archives and storing it in a central database. The purpose of indexing metadata is to make it available to you for searching.
- **EOSC**: European Open Science Cloud. Has the goal to develop “Web of FAIR Data and services’ for science in Europe. EOSC will be a multi-disciplinary environment where researchers can publish, find and re-use data, tools and services, enabling them to better conduct their work.

RESOURCES

- European Commission
  - Model agreement
  - Annotated grant agreement
  - Standard application form
  - DMP template (Templates & Forms > Project Reporting Templates) and via DMPonline.be
  - Programme Guide
  - ERC RDM and DMP guide

- Documentation on RDM, DMPs and FAIR data:
  - Knowledge clip on FAIR
  - Research tips: RDM requirements, OS requirements
  - H2020 DMP examples, at CORDIS website
    - https://cordis.europa.eu/project/id/776045/results
    - https://cordis.europa.eu/project/id/773649/results
    - https://www.assembleplus.eu/sites/assembleplus.eu/files/public/images_AquaTT/ASSEMBLE%20Plus%20Deliverable%20D2.10_v2_30.07.2018.pdf
  - RDM Guide of Elixir
  - FAIR principles
  - FAIR aware tool
  - FAIRsFAIR Data Object Assessment Metrics
  - FAIR cookbook
  - RDM kit of Elixir

BIBLIO DEMO

https://backoffice.biblio.ugent.be/dashboard

https://onderzoektips.ugent.be/en/tips/00002055/

UNIVERSITEIT
GENT

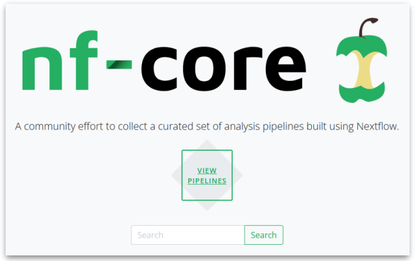
A short side note on computational workflows

Where can I find workflows?

Platform/System specific repositories

Galaxy PROJECT

IWC - Intergalactic Workflow Commission



Snakemake workflow catalog

UNIVERSITEIT GENT

The Git based repositories

GitHub

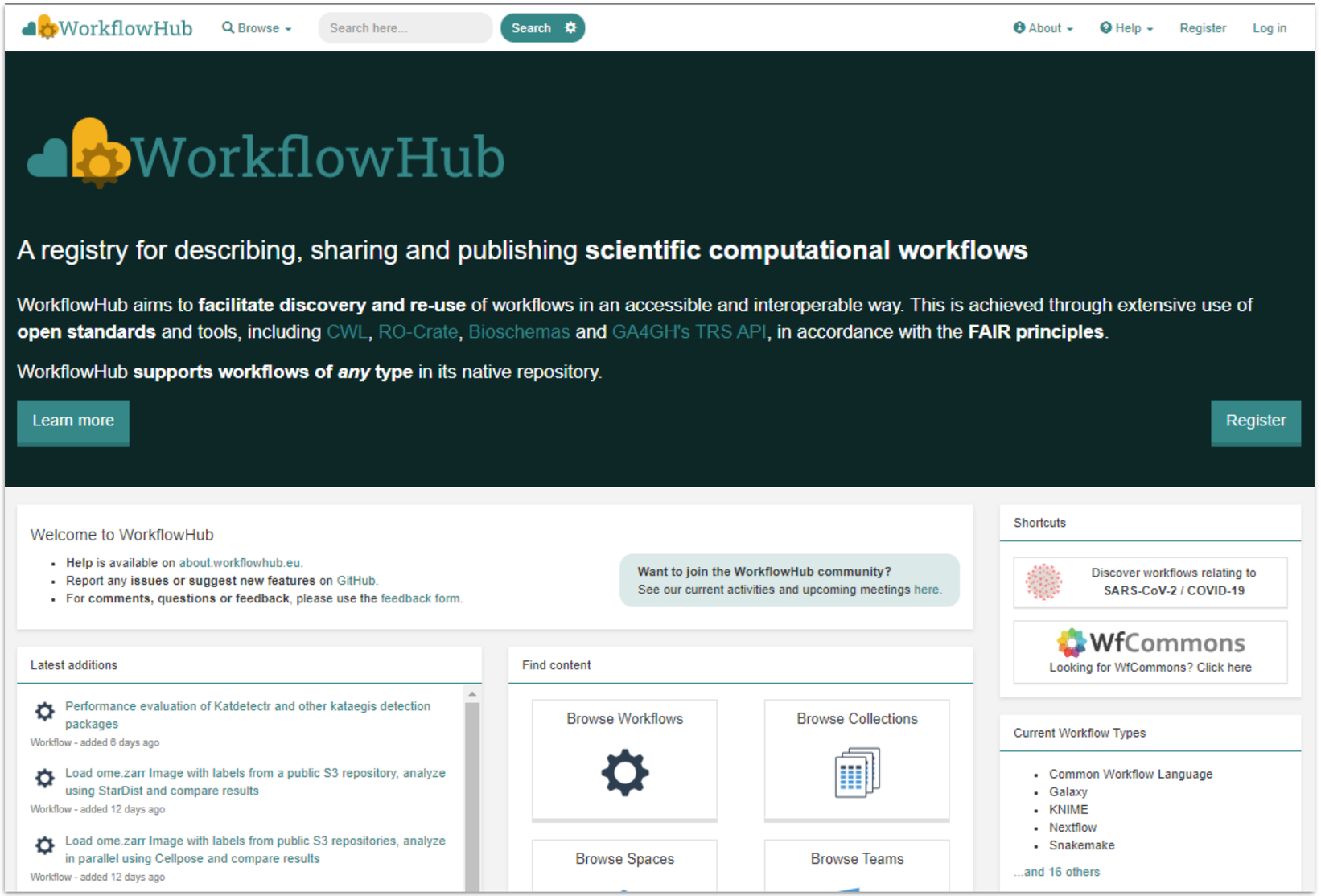
Generalist data repositories

zenodo

elixir BELGIUM

Using a registry to overcome fragmentation

Start: April 2020
Launch: Sept 2022



https://workflowhub.eu/